



University of Wisconsin
SCHOOL OF MEDICINE
AND PUBLIC HEALTH

Tutorial on Topological Data Analysis

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Abstract

Topological data analysis (TDA) provides a mathematical framework for extracting multiscale geometric and topological structure from complex data. Unlike conventional methods designed for Euclidean data, TDA naturally analyzes point clouds, networks, images, and other non-Euclidean objects through persistent homology, graph filtrations, and related topological invariants. This tutorial introduces the statistical foundations of TDA, including simplicial complexes, filtrations, persistent homology, persistence diagrams, and topological distances, together with practical computation using the standalone open-source MATLAB toolbox PH-STAT (<https://github.com/laplacebeltrami/PH-STAT>). Hands-on examples will cover structural and functional brain connectomics, dynamic brain networks, and other non-Euclidean datasets. The tutorial will also introduce recent advances in topological analysis of directed interactions and causal modeling, including Hodge-theoretic representations of feedback loops and cyclic information flow. Participants will gain both the theoretical background and practical tools needed to apply topological methods to modern data analysis problems.

Lecture materials for TDA tutorial

Standalone TDA matlab toolbox (50+ functions)

<https://github.com/laplacebeltrami/PH-STAT>

Tutorial slides are in this site.

Plan to have 200+ functions

Detailed software manual with basic theory (49 pages). Plan to have 300+ page manual

<https://arxiv.org/abs/2304.05912>

Lecture slides have [unlerlined links](#) for additional information and papers.

Where topology came from?

Random
field

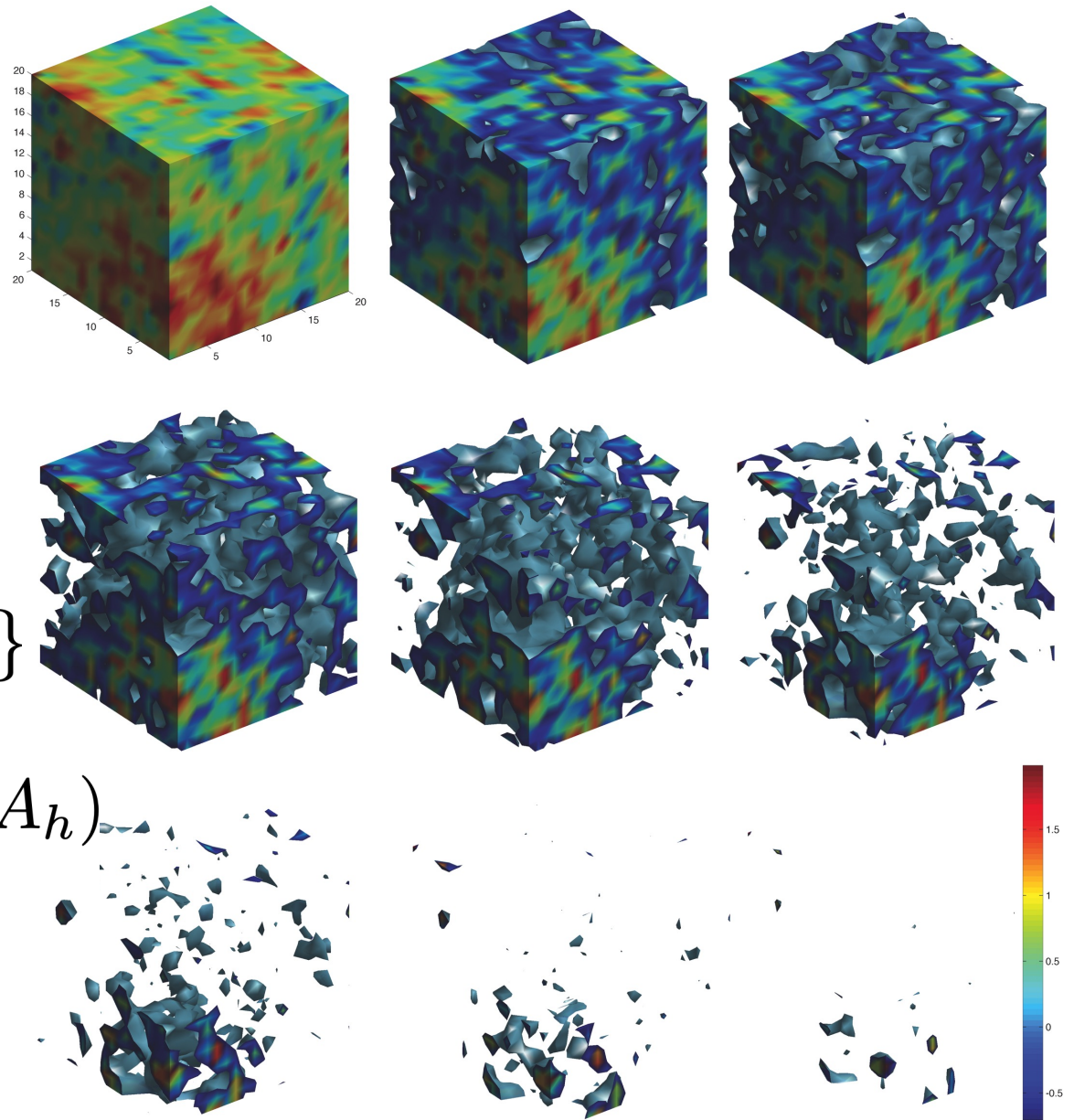


$$P\left(\sup_{x \in \mathbb{M}} T(x) > h\right)$$

$$A_h = \{x \in \mathbb{M} : T(x) > h\}$$

$$P\left(\sup_{x \in \mathbb{M}} T(x) > h\right) = \mathbb{E}\chi(A_h)$$

$$\chi(A_h) = \sum_j (-1)^j \beta_j(A_h)$$



Milnor 1963 Morse theory

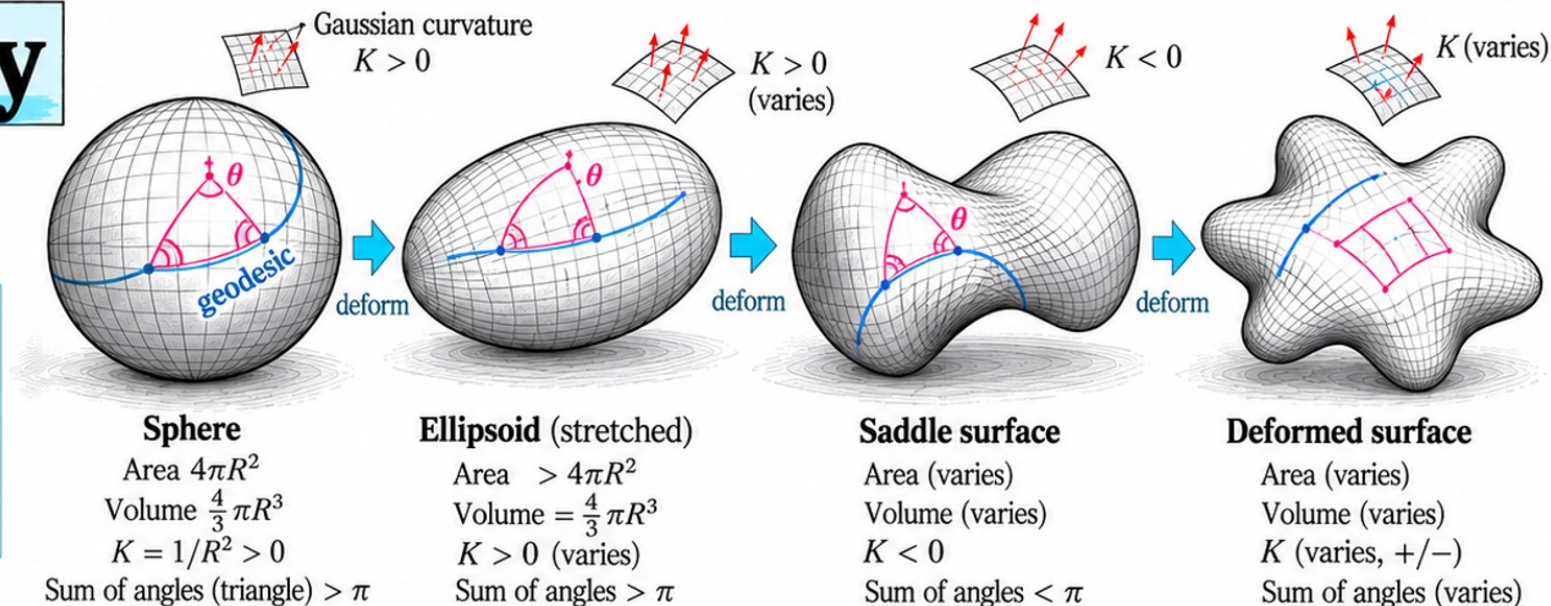
Adler, 1994 The geometry of random fields

Worsley et al., 1996 Human Brain Mapping

Geometry

Quantities that **change** under continuous deformation.

- length (geodesic distance)
- angle
- curvature
- area / surface area
- volume
- metric, coordinates, ...

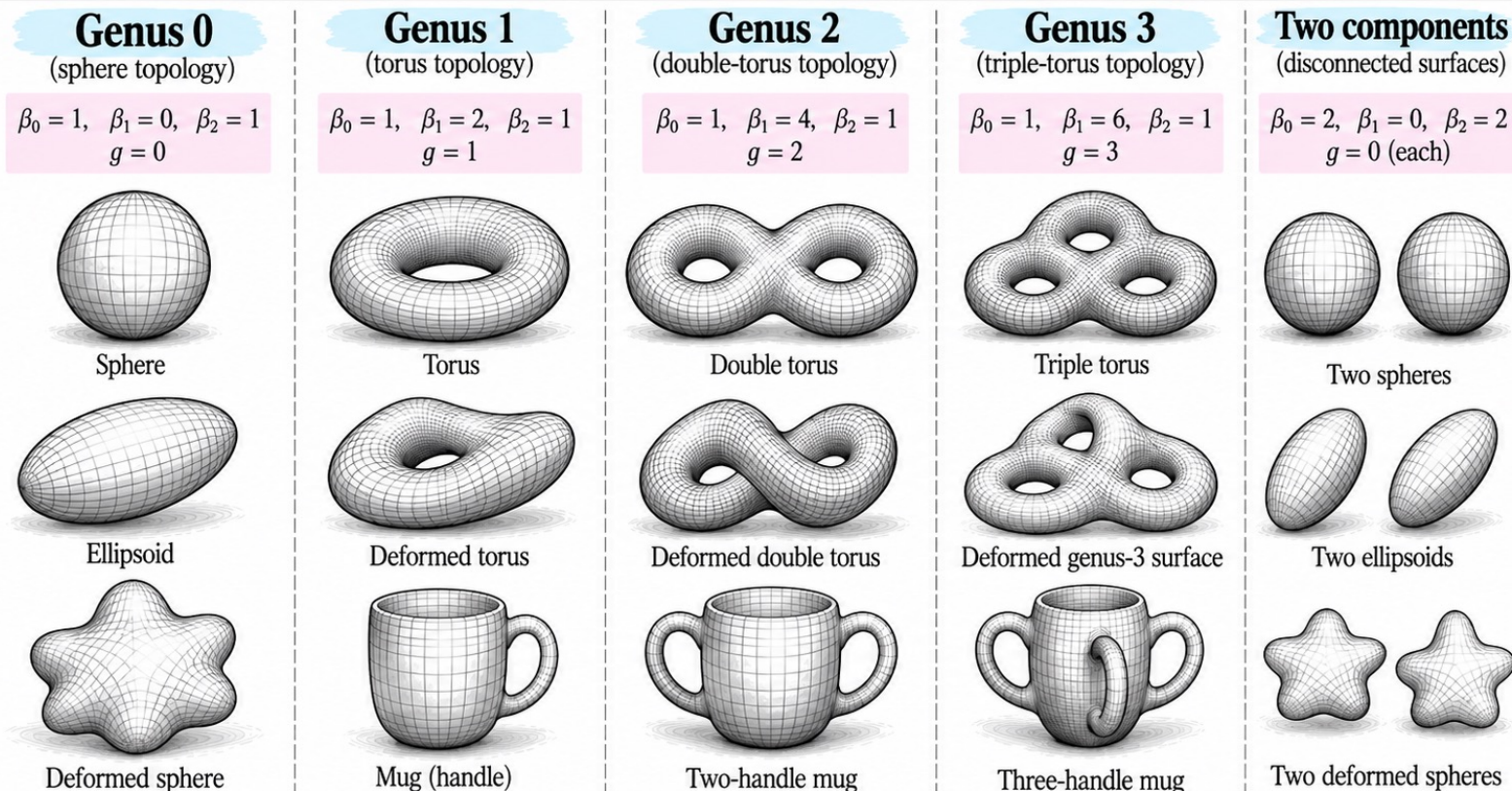


Topology

Quantities that **remain invariant** under continuous deformation.

- number of connected components β_0
- number of holes (1D) β_1
- number of voids (2D) β_2
- genus g
- cannot change without cutting or gluing

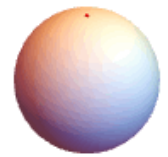
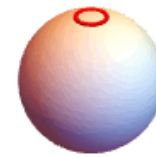
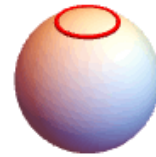
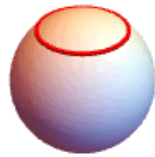
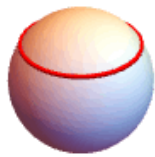
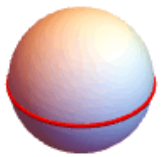
Shapes in the same column are topologically equivalent (they can be continuously deformed into each other).



Homology studies which objects cannot be continuously contracted to a lower-dimensional object (ultimately a point).

H_0 Connected components. Every connected component contracts to a point.

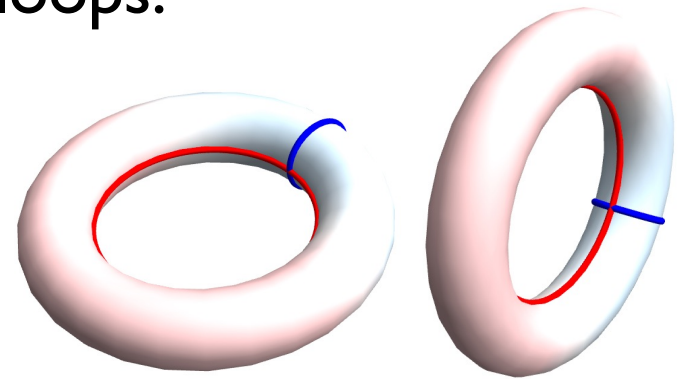
Betti number $\beta_0 = \dim(H_0)$ counts the resulting points.



$$\beta_0 = 1, \beta_1 = 0, \beta_2 = 1$$

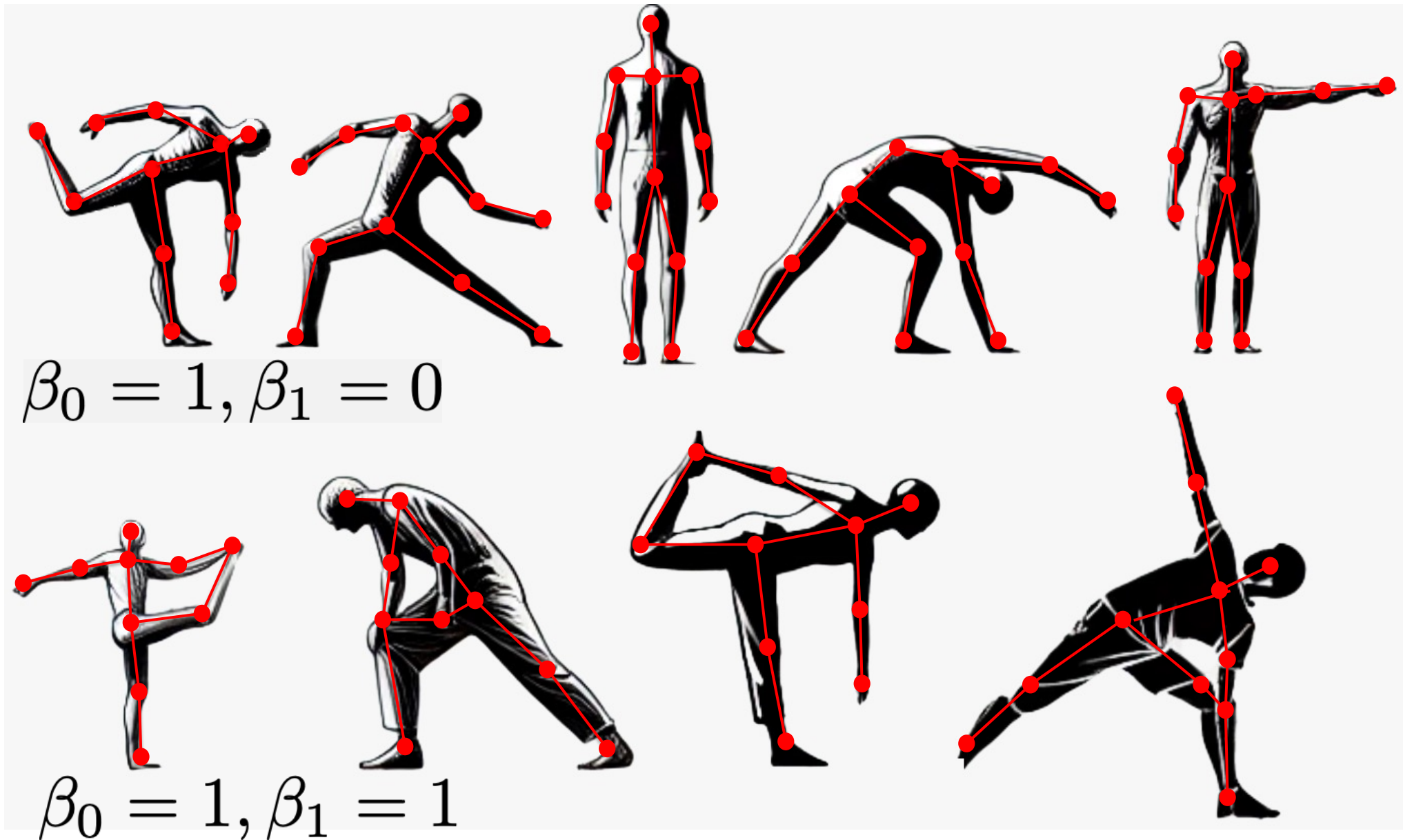
H_1 Loops that cannot be contracted to a point
 $\beta_1 = \dim(H_1)$ counts the resulting loops.

H_2 Closed surfaces that cannot be contracted to a point because they enclose a cavity.



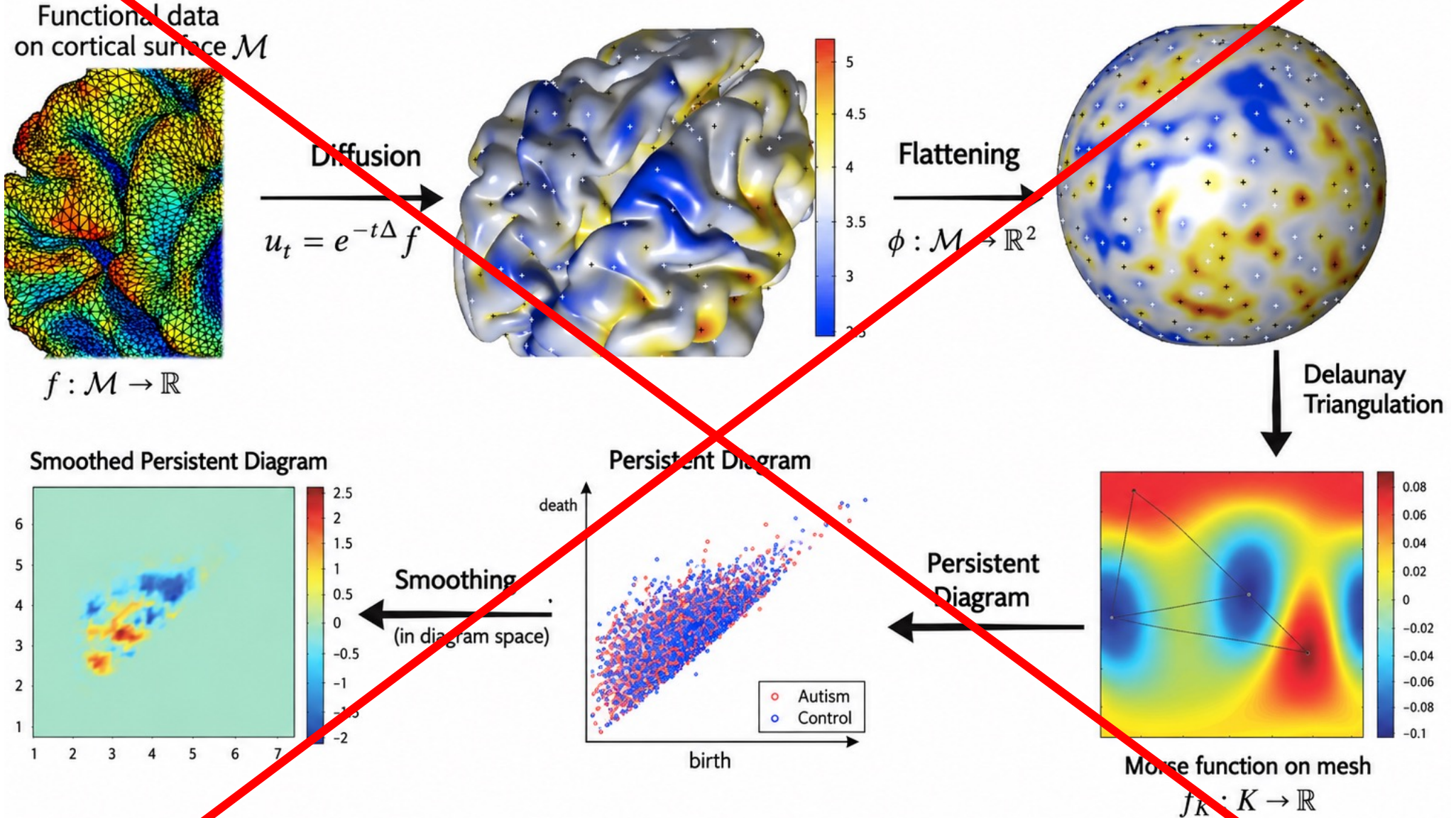
$$\beta_0 = 1, \beta_1 = 2, \beta_2 = 1$$

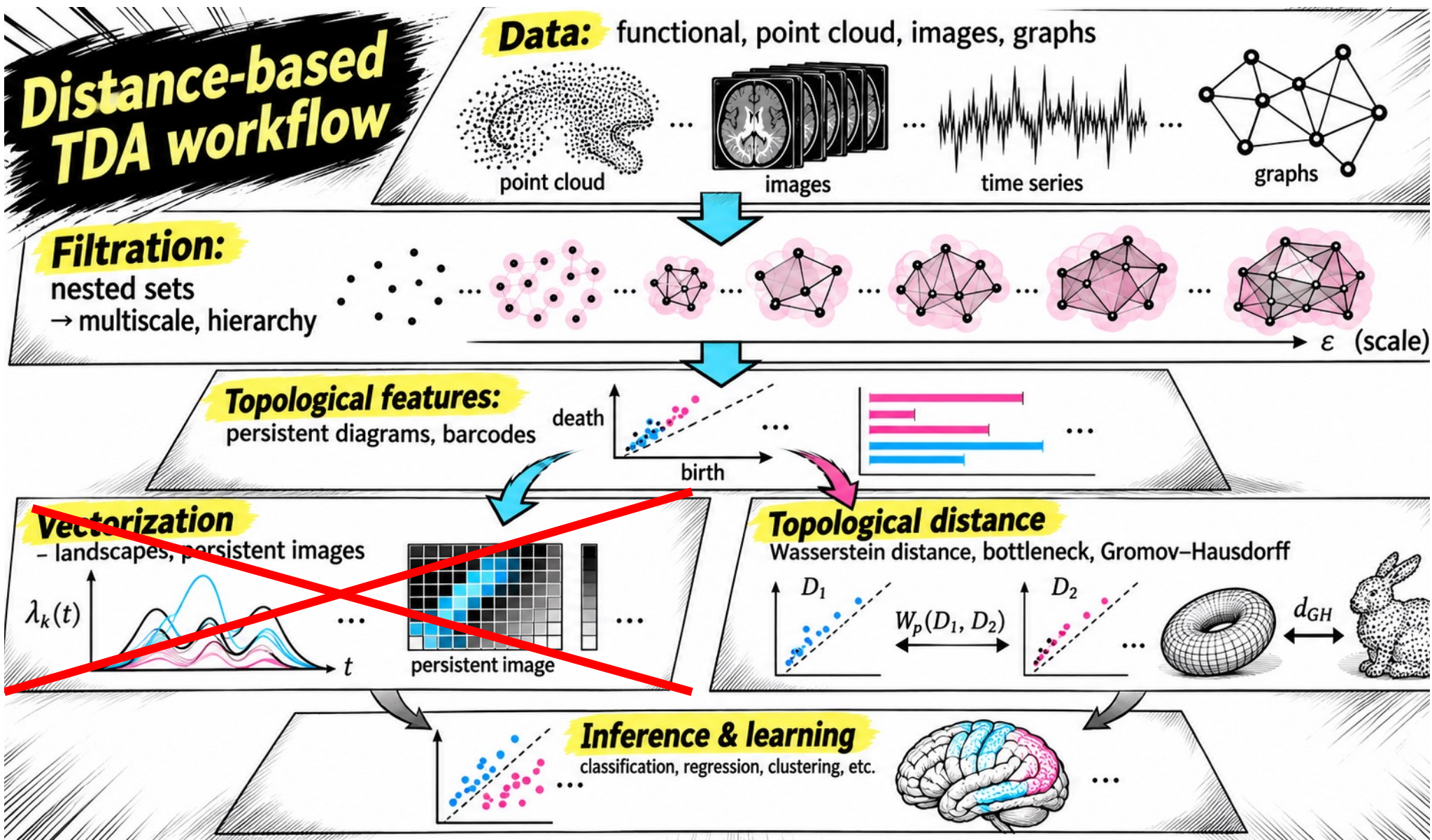
Topological variability is zero.



Can you come up with a method that gives 0 variability?

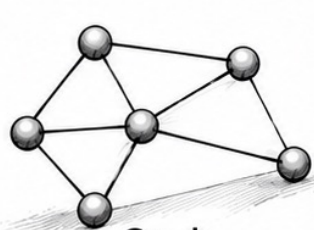
Morse filtration on functional data



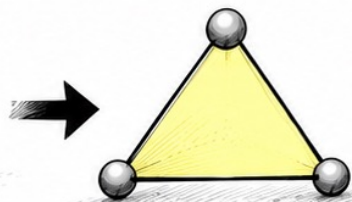


Simplicial complex Data Structure

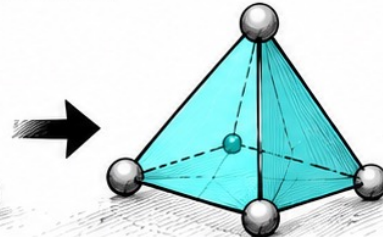
A **graph** represents only pairwise interactions through vertices and edges.
A **simplicial complex** extends a graph by including higher-order structures such as filled triangles (2-simplices) and filled tetrahedra (3-simplices), enabling the representation of multi-way interactions.



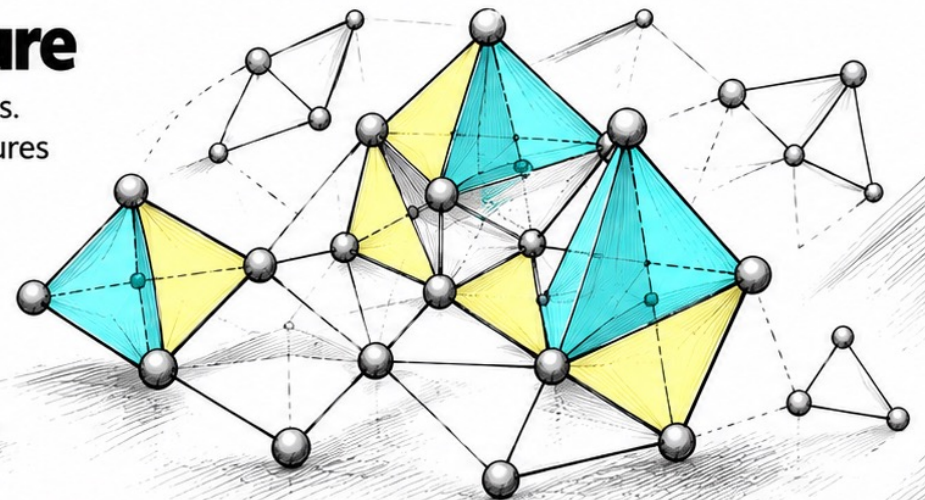
Graph
(vertices + edges)



2-simplex
(filled triangle)



3-simplex
(filled tetrahedron)



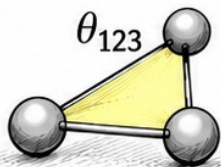
Simplicial complex

Why simplicial complex?

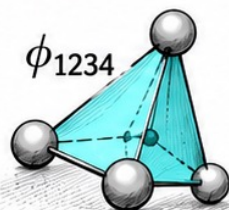
Encoding **higher order interactions** in protein structure



2-atom interaction



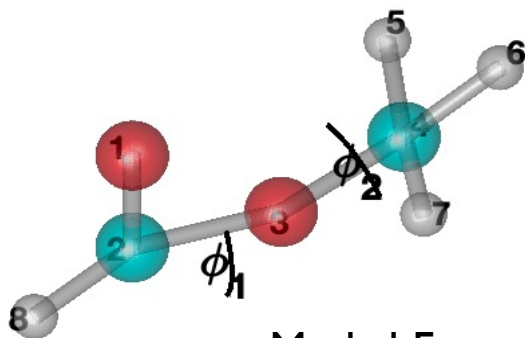
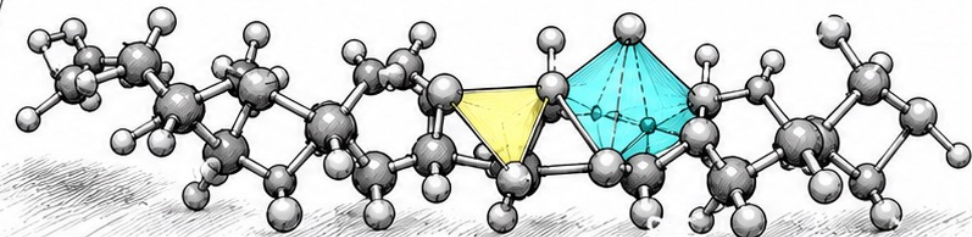
3-atom interaction
(3 atoms form a plane)



4-atom interaction
(dihedral angle)

Potential energy

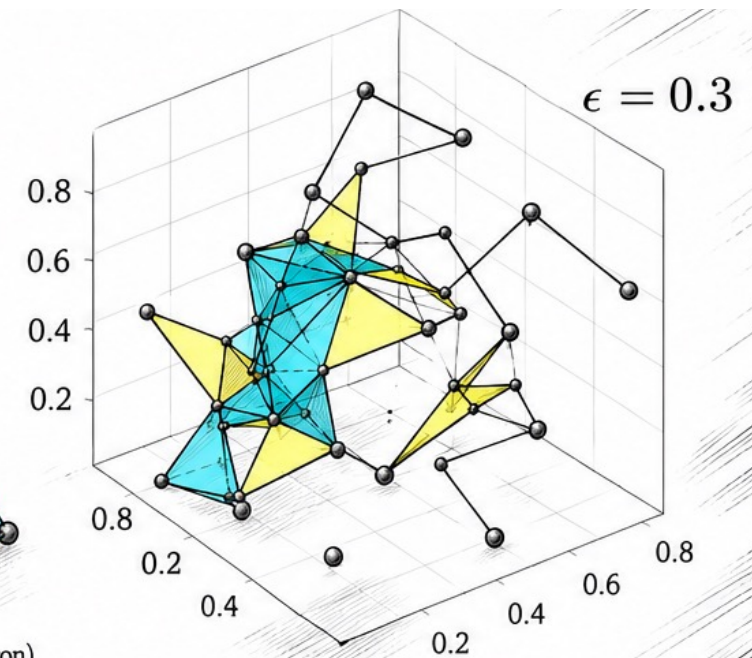
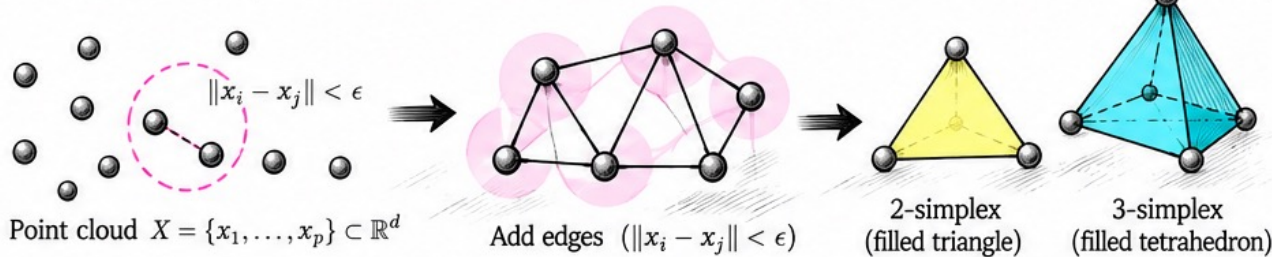
$$V = \underbrace{\sum_{(i,j) \in B} V_2(r_{ij})}_{\text{function over edges}} + \underbrace{\sum_{(i,j,k) \in A} V_3(\theta_{ijk})}_{\text{function over triangles}} + \underbrace{\sum_{(i,j,k,l) \in D} V_4(\phi_{ijkl})}_{\text{function over tetrahedra}}$$



Methyl Formate

Rips complex

Baseline simplicial complex is the **Rips complex** constructed by connecting pairs of points whose distance is less than radius ϵ . Whenever all edges of a simplex are present, its interior is filled automatically.



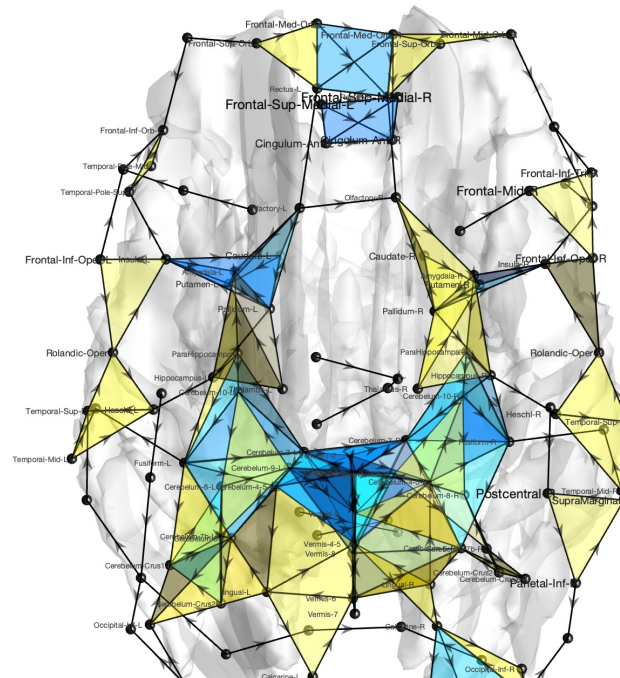
Matlab implementation in PH-STAT

```
p = 50; d = 3;
X = rand(p, d);

k = 3;
radius = 0.3;
S = PH_rips(X, k, radius);
PH_rips_display(X, S);
```

S = 4x2 cell array

```
{50×1 double}    % vertices (0-simplices)
{101×2 double}   % edges (1-simplices)
{73×3 double}    % triangles (2-simplices)
{27×4 double}    % tetrahedra (3-simplices)
```



Connectivity

 $S\{3, 1\}$ (triangles)
$$\begin{bmatrix} 1 & 5 & 13 \\ 1 & 5 & 16 \\ 1 & 5 & 26 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Functional data over simplex

 $S\{3, 2\}$
$$\begin{bmatrix} 0.2932 \\ 0.2579 \\ 0.2766 \\ \dots \end{bmatrix}$$
 $S\{4, 1\}$ (tetrahedra)
$$\begin{bmatrix} 2 & 7 & 18 & 24 \\ 3 & 5 & 12 & 20 \\ 3 & 6 & 14 & 28 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$
$$S\{4, 2\}$$
$$\begin{bmatrix} 0.1254 \\ 0.3187 \\ 0.2191 \\ \dots \end{bmatrix}$$

Directed interactions

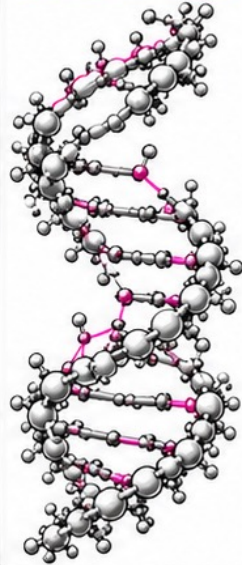
Rips filtration

DNA double helix

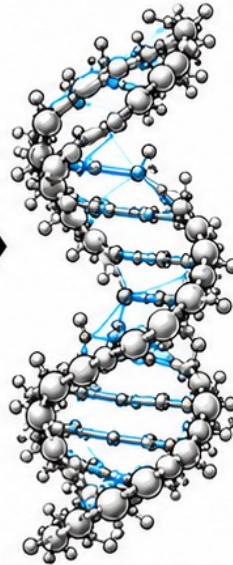
Atomic point cloud
 $\mathcal{X} = \{x_i\} \subset \mathbb{R}^3$

Rips filtration: connect points within distance ϵ

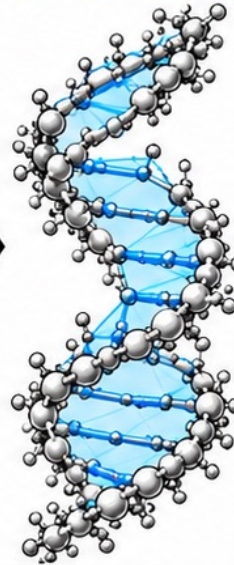
$\epsilon = 1.5 \text{ \AA}$



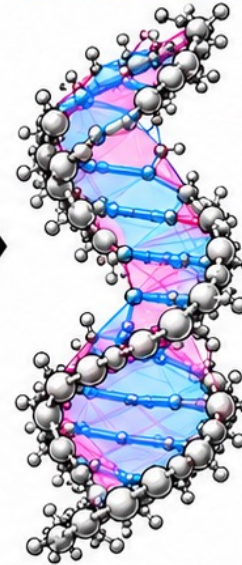
$\epsilon = 2.5 \text{ \AA}$



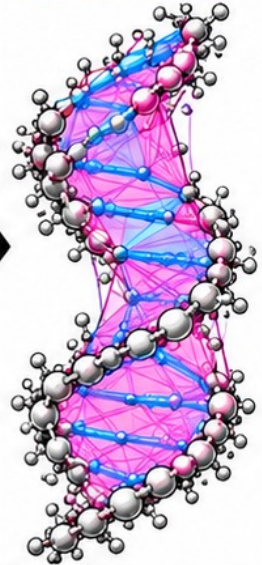
$\epsilon = 3.5 \text{ \AA}$



$\epsilon = 4.5 \text{ \AA}$



$\epsilon = 6.0 \text{ \AA}$

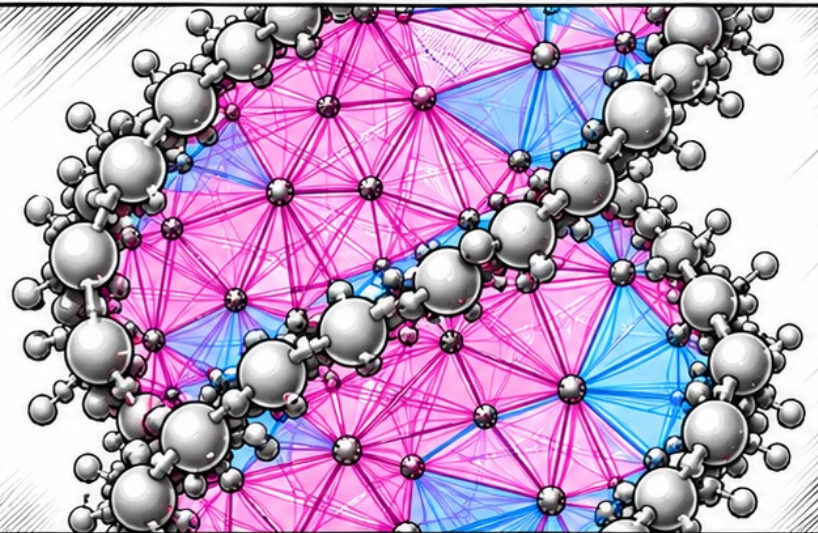


Limitation:
Dense complete graph

Computational complexity:

$$O(n2^n)$$

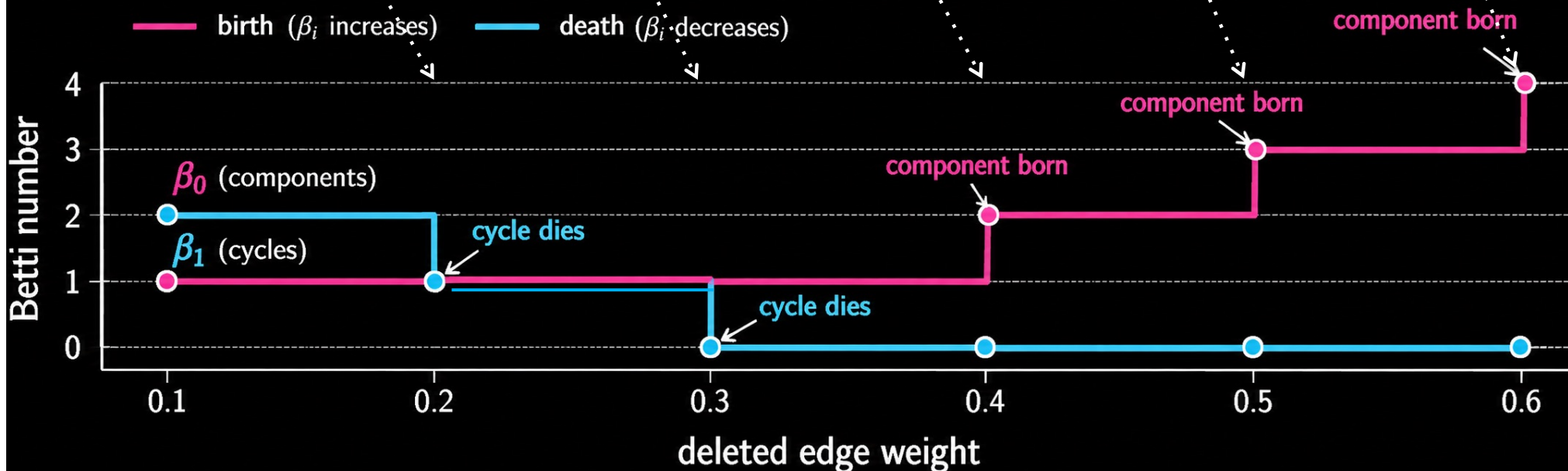
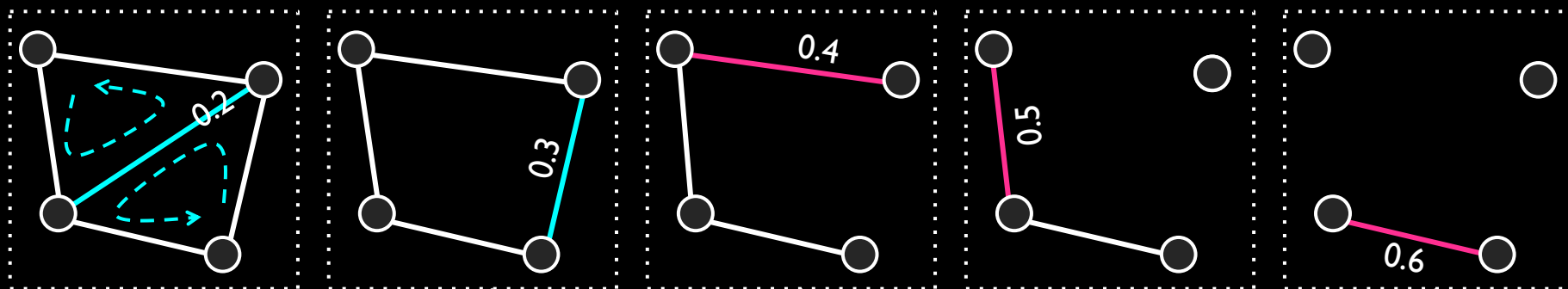
n : number of points (atoms)



Graph filtration (edge deletion from smallest to largest)

Deleting edges destroy cycles

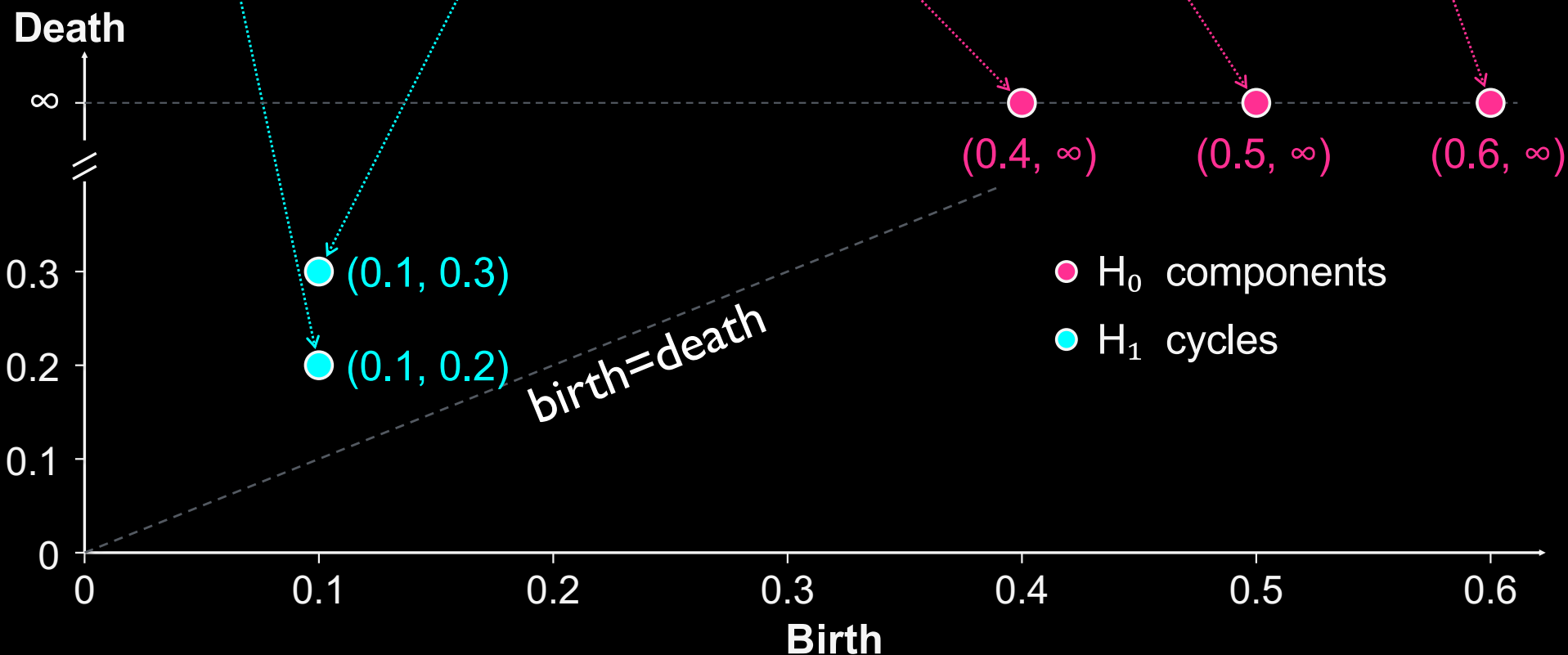
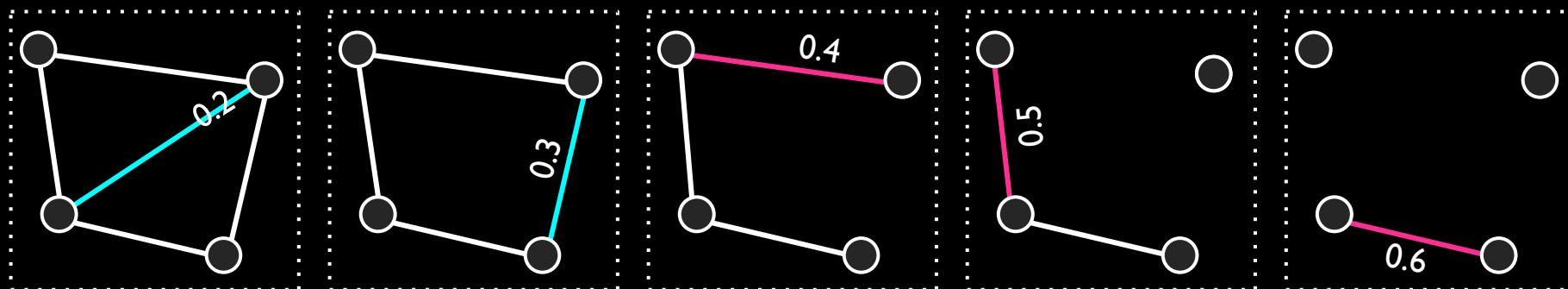
Deleting edges create components



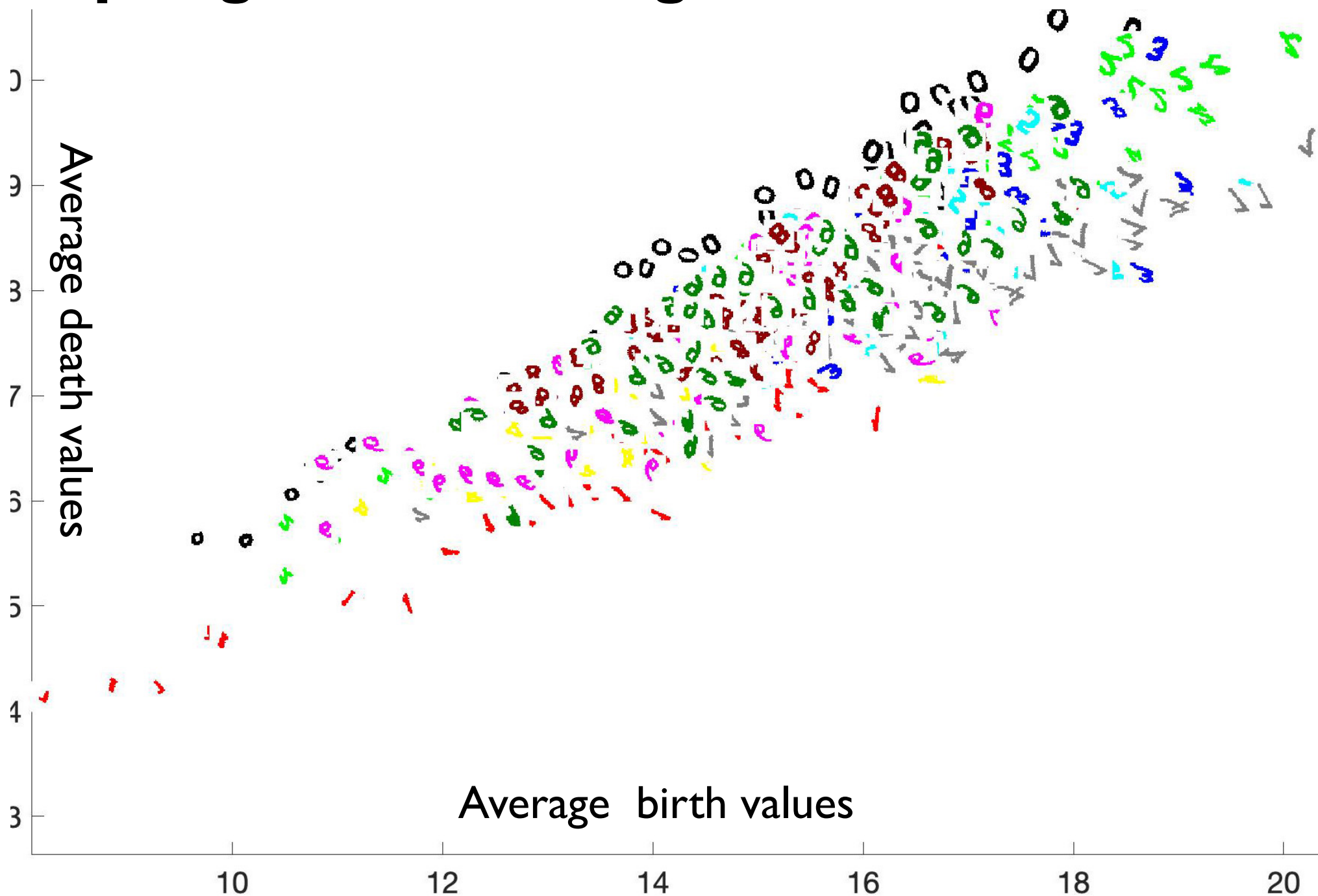
Persistent diagram (inefficient representation)

Deleting edges destroy cycles

Deleting edges create components

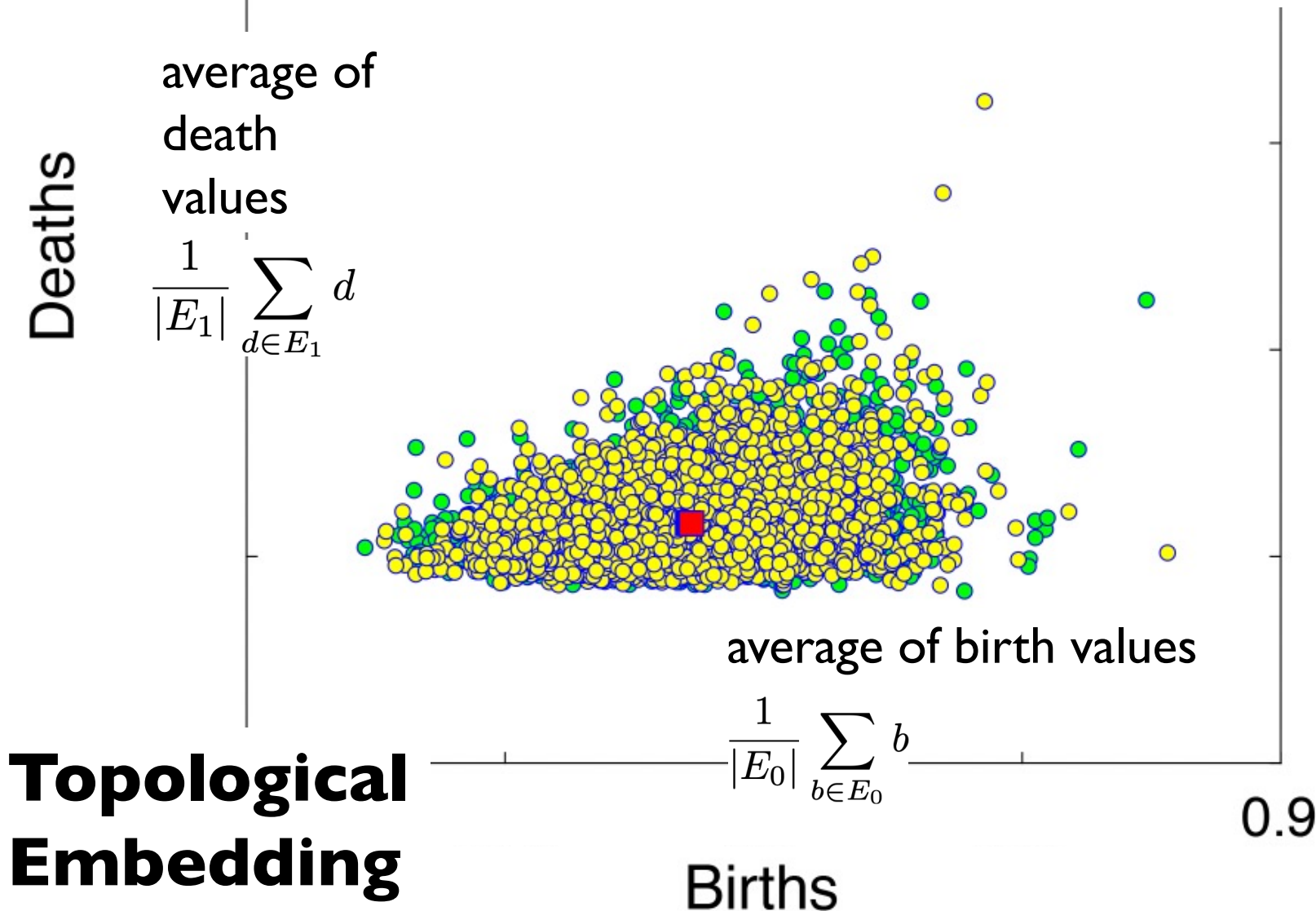


Topological Embedding on MNIST

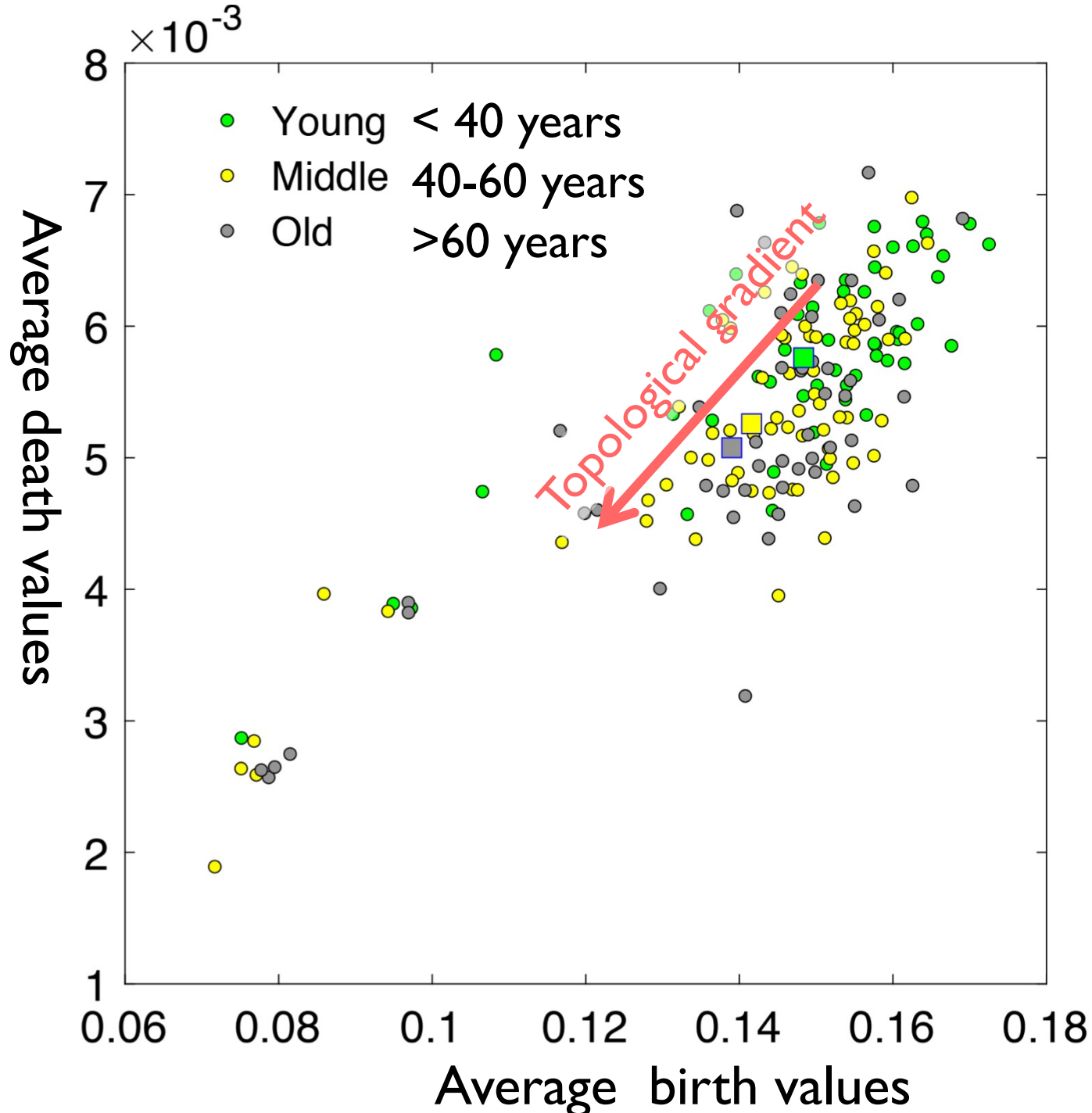


Covariate-free Method

Male and female differences are mostly geometric.



Topological Embedding of Structural Connectivity (dMRI)



Data from

Jeffrey Phillips

James Gee

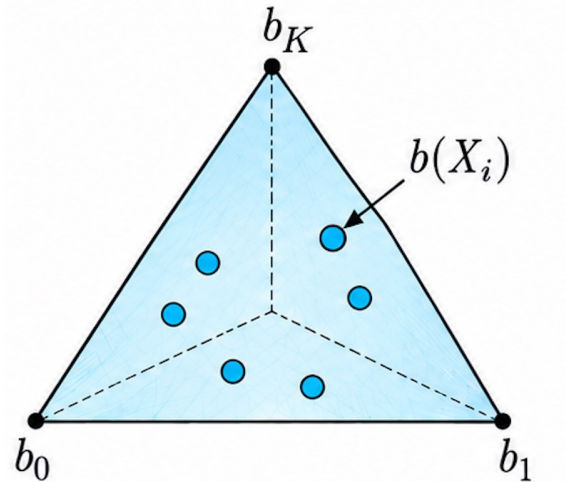
Univ. of Pennsylvania

Aging is topological.

Topological distance $\rightarrow$ Euclidean distance

Birth polytope $\mathcal{B}$

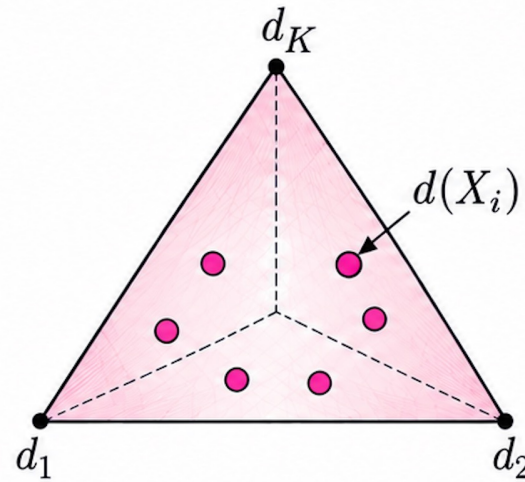
$$b_0 < b_1 < \dots < b_K$$



$$\mathcal{B} = \{b \in \mathbb{R}^{K+1} : b_0 < b_1 < \dots < b_K\}$$

Death polytope $\mathcal{D}$

$$d_1 < d_2 < \dots < d_K$$

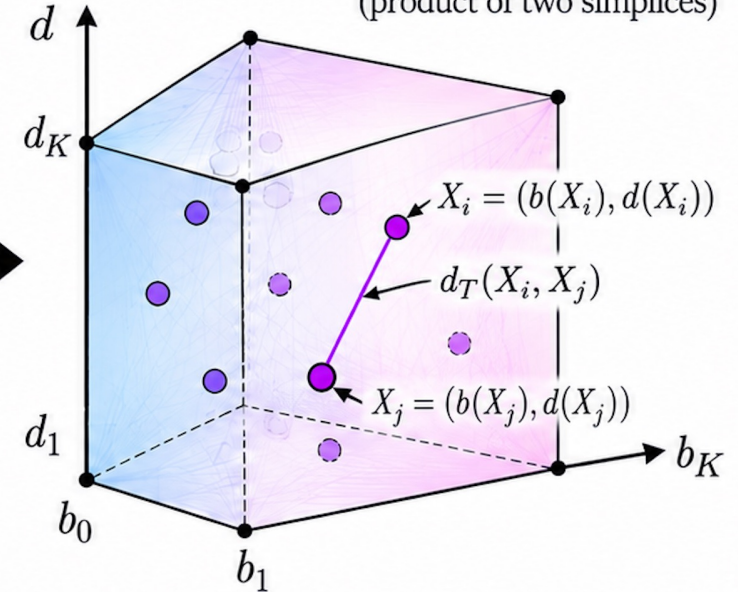


$$\mathcal{D} = \{d \in \mathbb{R}^K : d_1 < d_2 < \dots < d_K\}$$

Product polytope $\mathcal{P} = \mathcal{B} \times \mathcal{D}$

$$\mathcal{P} = \{(b, d) : b \in \mathcal{B}, d \in \mathcal{D}\}$$

(product of two simplices)



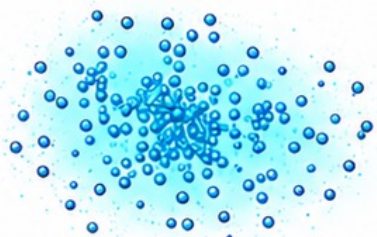
$$d_T(X_i, X_j)^2 = \|b(X_i) - b(X_j)\|^2 + \|d(X_i) - d(X_j)\|^2$$

Run time $\mathcal{O}(q \log q)$ for q number of edges

Wasserstein distance

$$X \sim f_1$$

$$Y \sim f_2$$



$$P_1 = \{x_1, \dots, x_q\} \subset \mathbb{R}^2$$

$$P_2 = \{y_1, \dots, y_q\} \subset \mathbb{R}^2$$

$$f_1(x) = \frac{1}{q} \sum_{i=1}^q \delta(x - x_i)$$

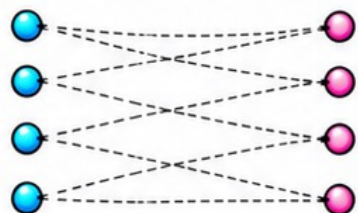
$$f_2(y) = \frac{1}{q} \sum_{i=1}^q \delta(y - y_i)$$

r -Wasserstein distance:

$$d(\mu, \nu)^r = \inf_{(X,Y)} \mathbb{E} \|X - Y\|^r$$

2-Wasserstein distance:

$$d(P_1, P_2) = \inf_{\psi: P_1 \rightarrow P_2} \left(\sum_{x \in P_1} \|x - \psi(x)\|^2 \right)^{1/2}$$

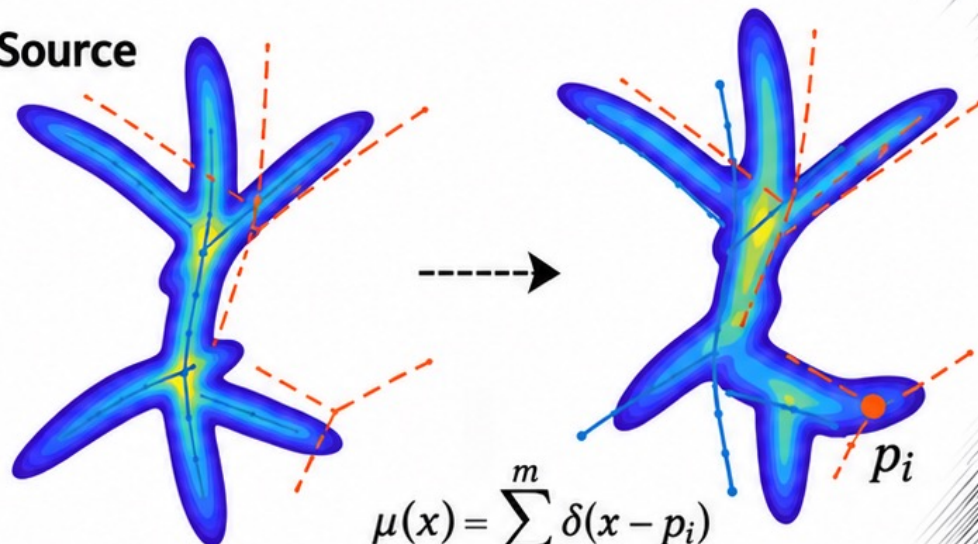


Assignment problem:
Hungarian algorithm
 $\mathcal{O}(q^3)$

Optimal bijection ψ^*

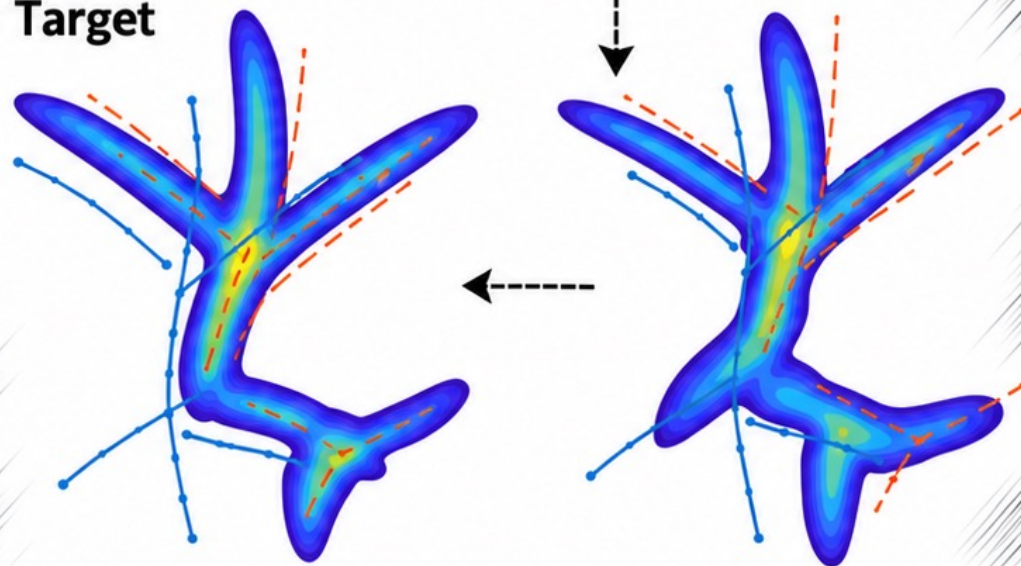
Wasserstein registration of trees

Source



$$\mu(x) = \sum_{i=1}^m \delta(x - p_i)$$

Target



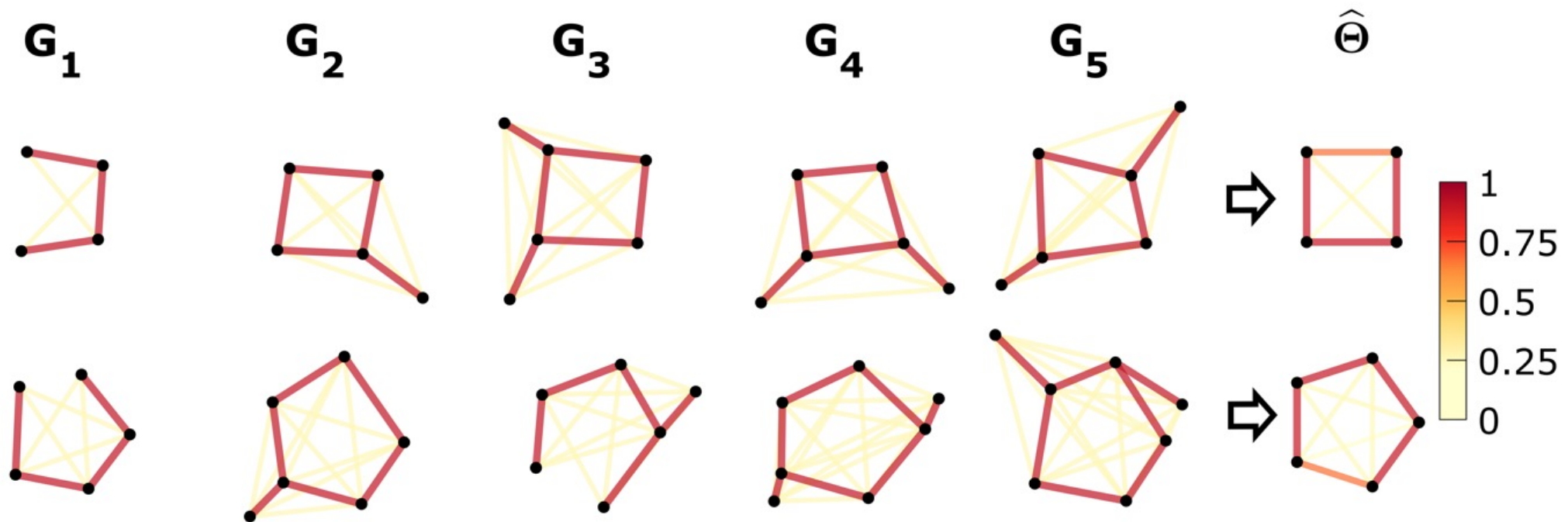
Poisson potential

$$\Delta u + \lambda u = \mu$$

Topological averaging

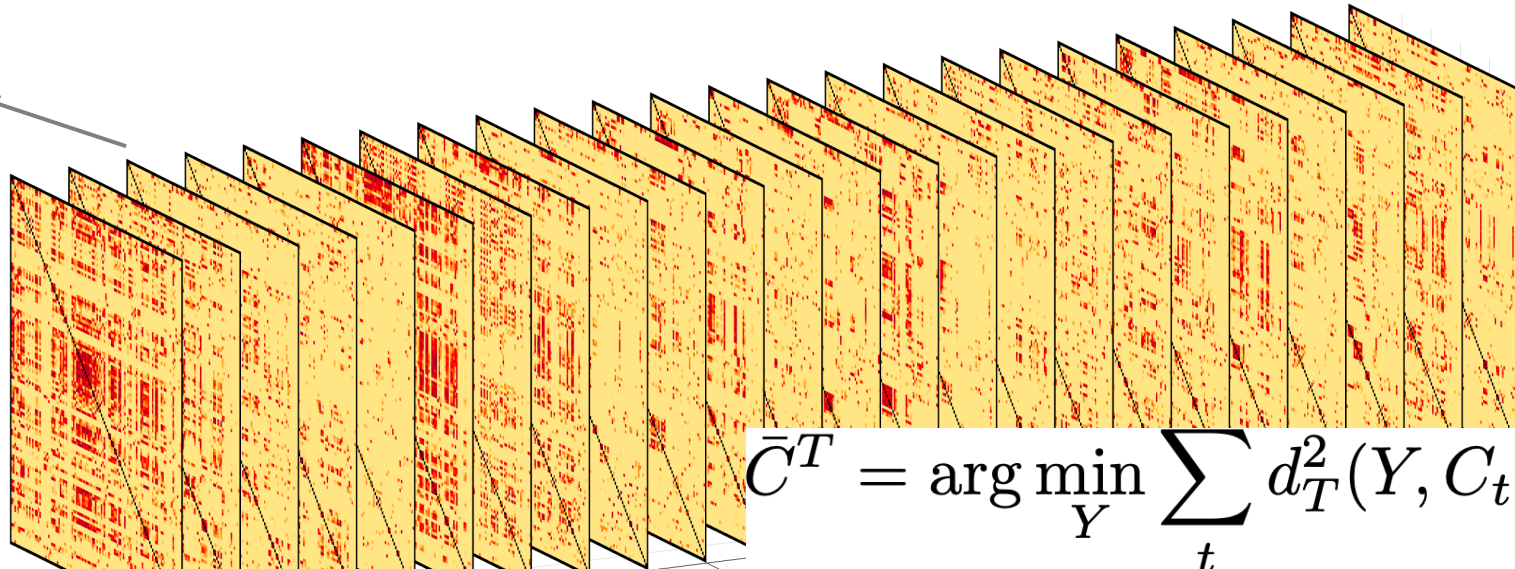
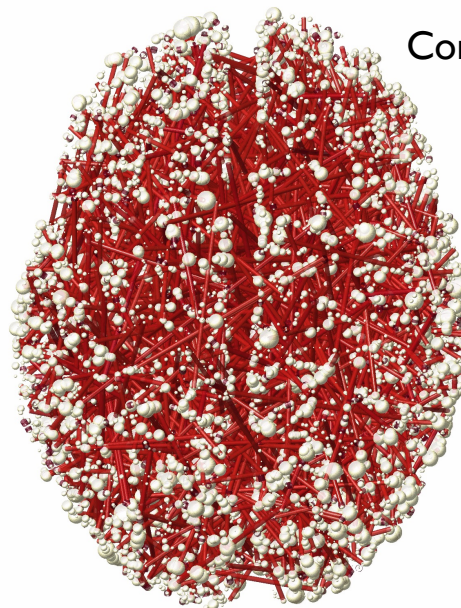
= minimizer of topological distance

$$\hat{\Theta} = \operatorname{argmin}_{\Theta} \sum_{i=1}^n d_T^2(G_i, \Theta)$$



We can average rs-fMRI across time and subjects (n=400, 15mn)

Correlation above 0.7

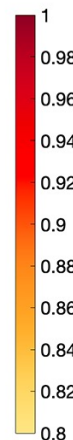
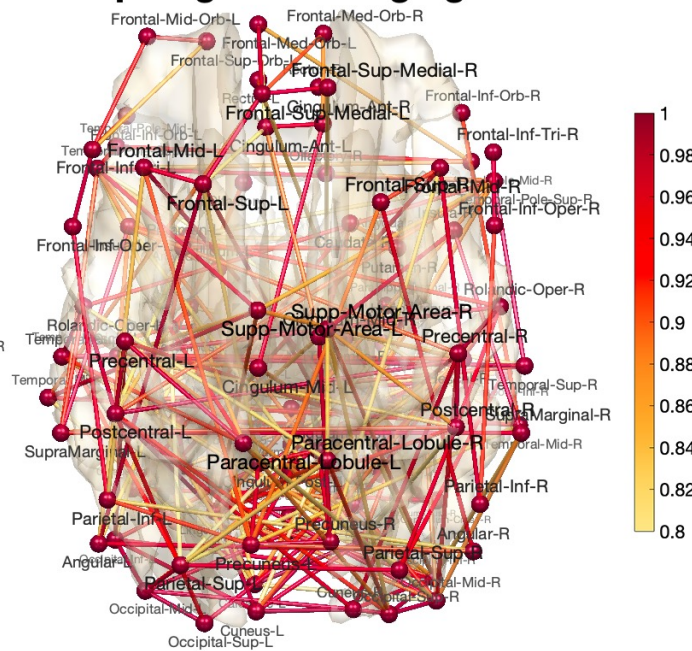
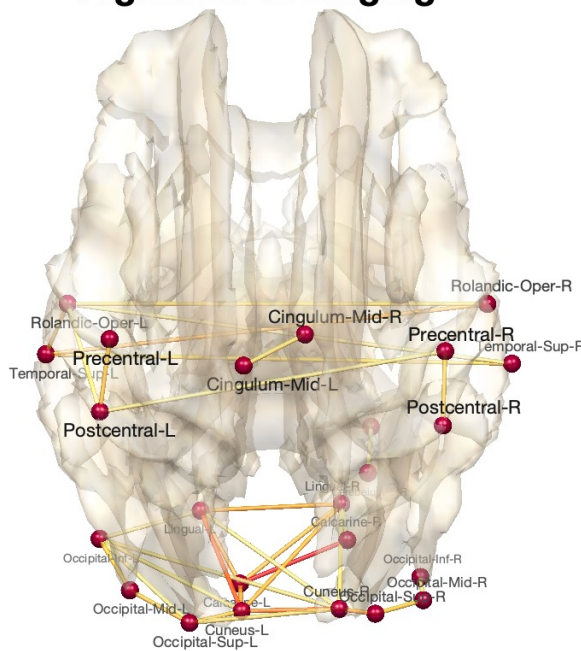
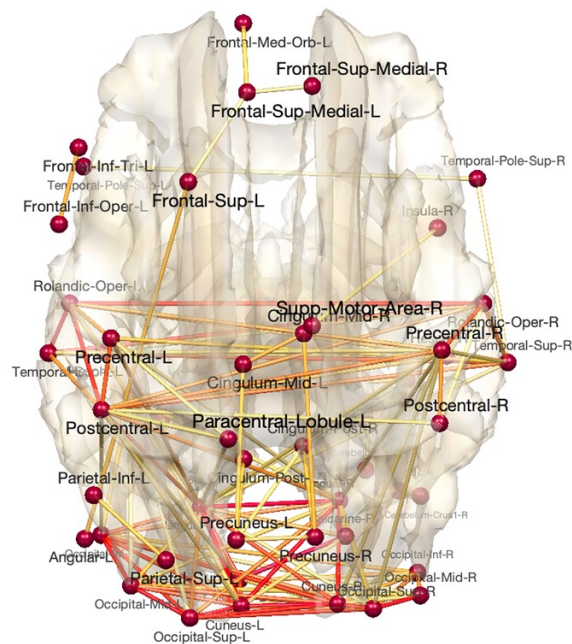


$$\bar{C}^T = \arg \min_Y \sum_t d_T^2(Y, C_t)$$

Correlation

Algebraic averaging

Topological averaging

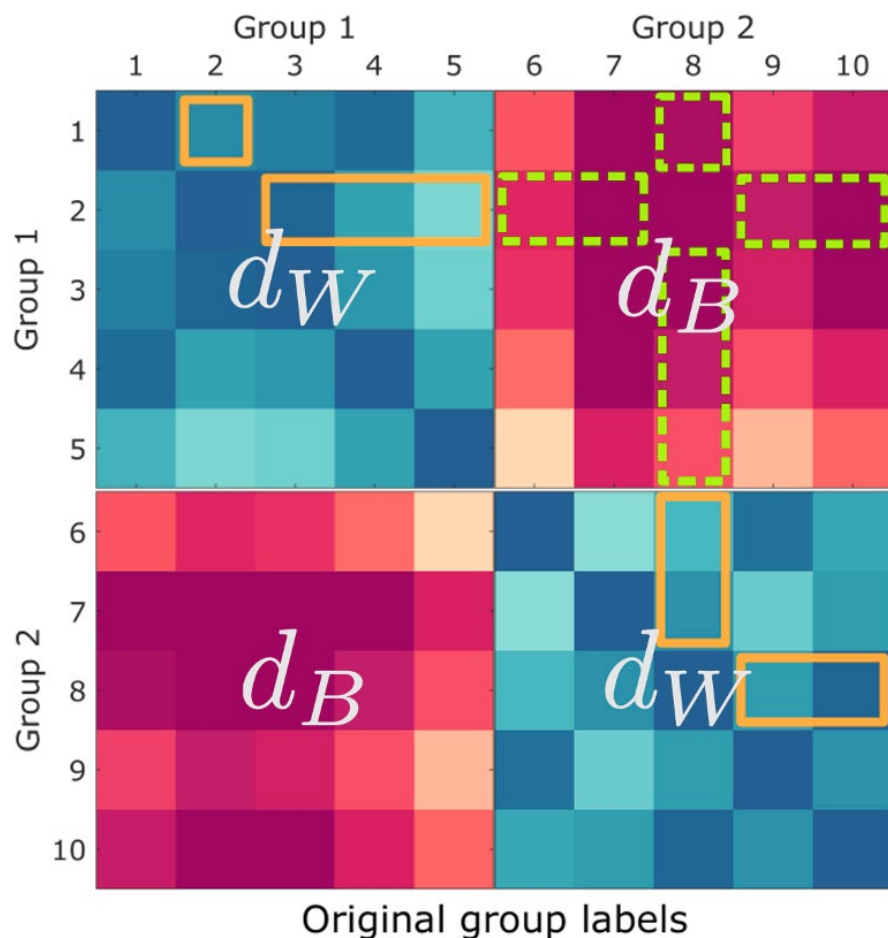


Sample variance

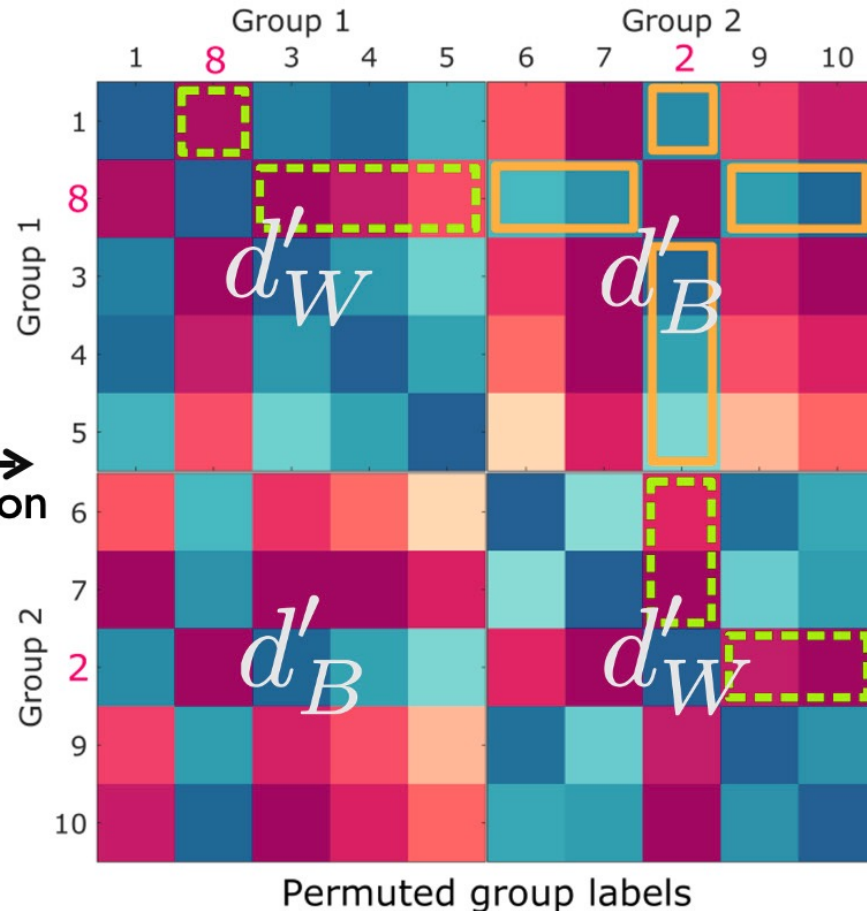
0.47+-0.02

0.02+-0.01

Topological inference via online permutation test



Transposition →

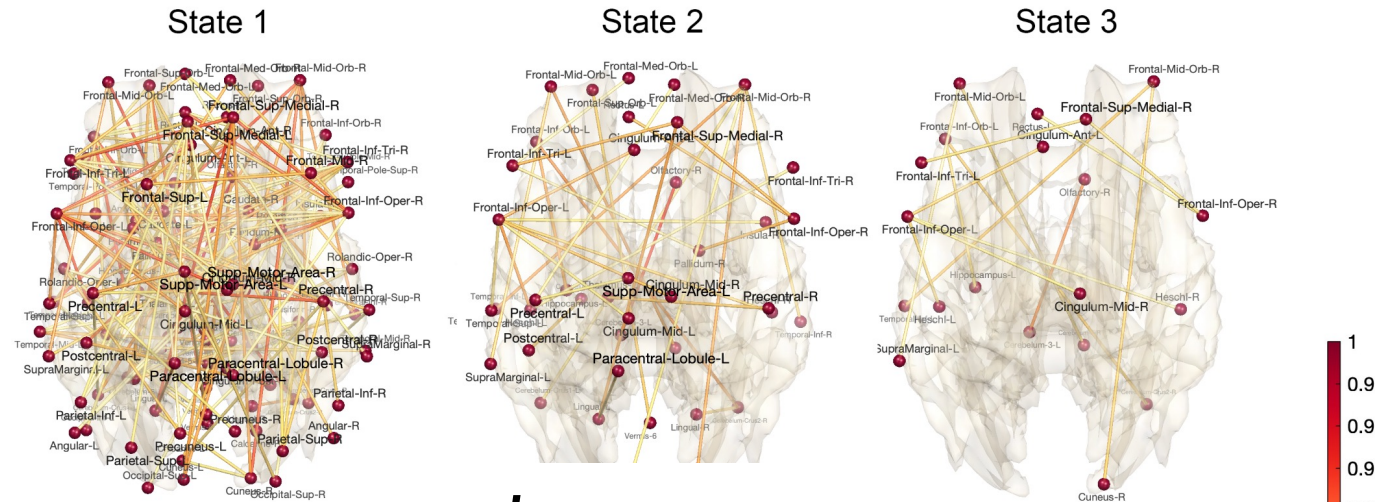


Distance matrix precomputed.
A permutation only rearrange terms

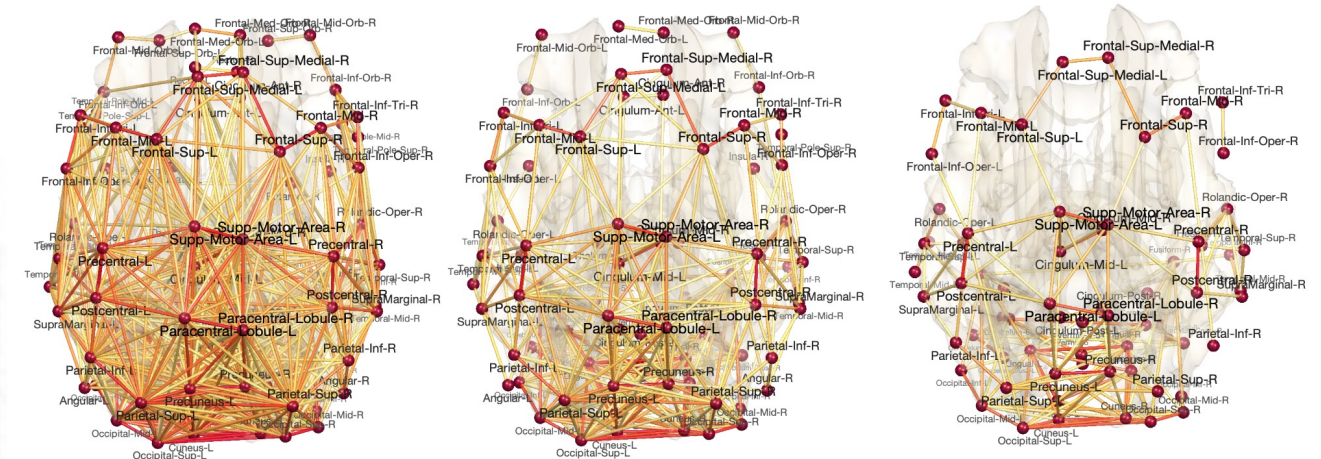
100 million
transpositions < 2 mins

Topological clustering

rs-fMRI connectivity
($n=479$, UW-Madison)

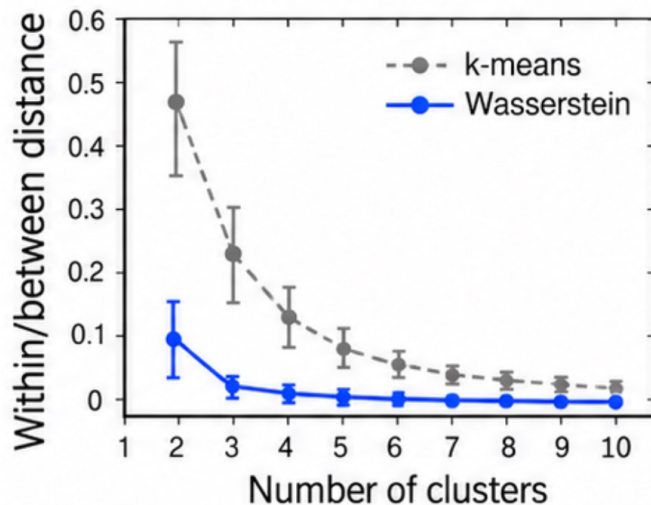


k-means



Wasserstein

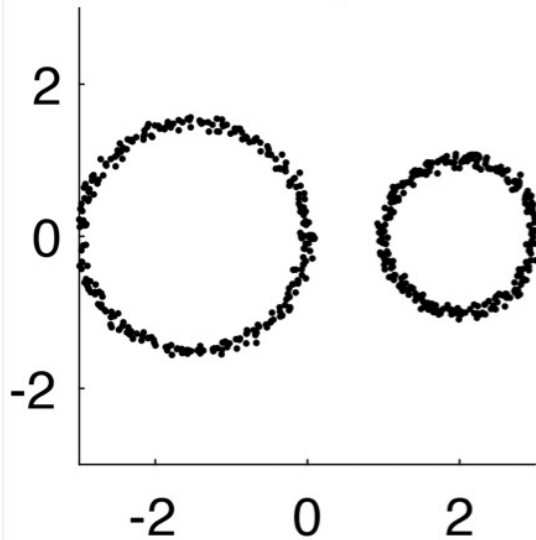
6 times smaller ratio



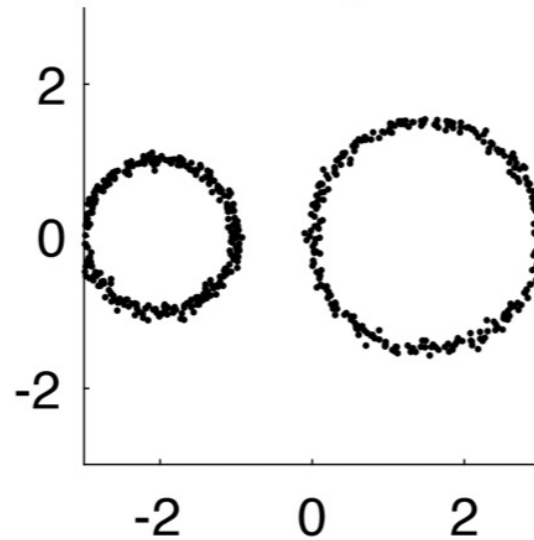
Chung et al. 2024 PLOS Computational Biology

Existing clustering fails in topological task

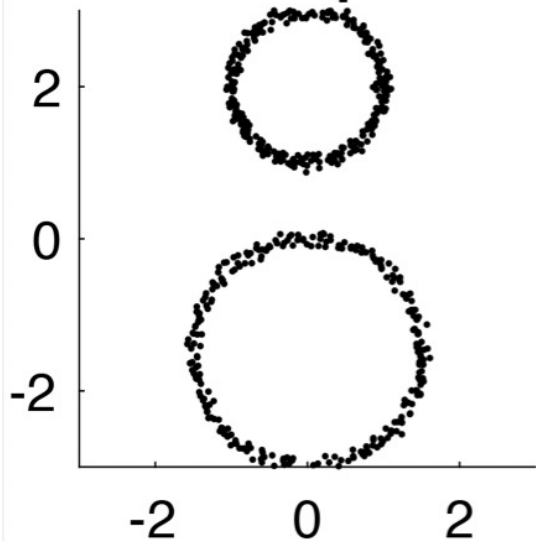
Group 1



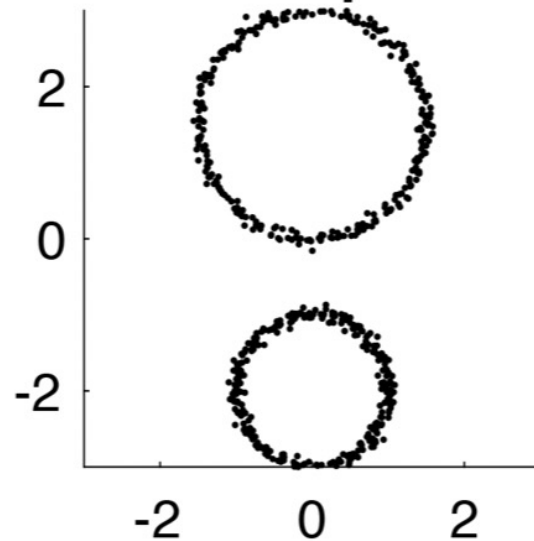
Group 2



Group 3



Group 4



K-means clustering 0.98 ± 0.01
Hierarchical clustering 1.00 ± 0.00



Topological clustering 0.63 ± 0.04

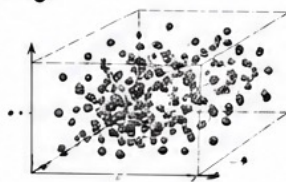
Spectral TDA workflow

Data:

functional, point cloud, images, graphs



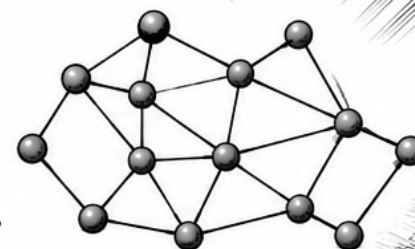
functional



point cloud



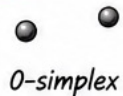
images



graphs

Simplicial complex:

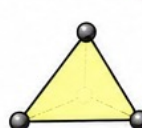
boundary matrices



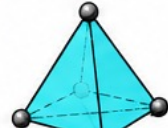
0-simplex



1-simplex

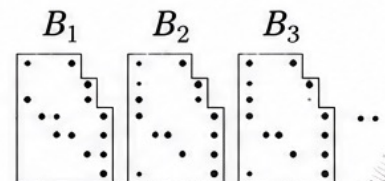


2-simplex



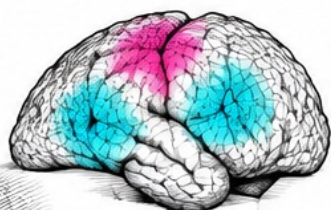
3-simplex

... $B_k : C_k \rightarrow C_{k-1}$

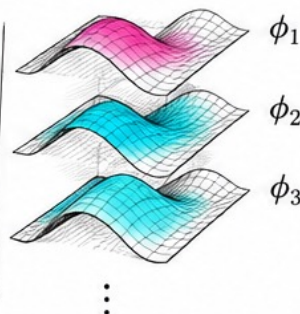


Hodge Expansion

Functional data analysis
Denoising, regression



$$f = \sum_i \alpha_i \phi_i$$



ϕ_1

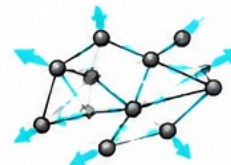
ϕ_2

ϕ_3

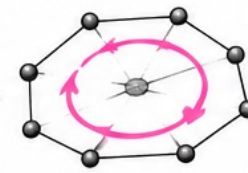
Hodge decomposition

Directed interactions,
cyclic interactions

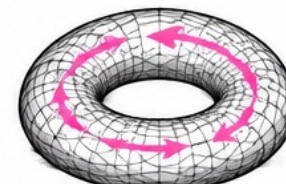
$$X = X_G + X_C + X_H$$



gradient (X_G)



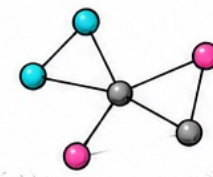
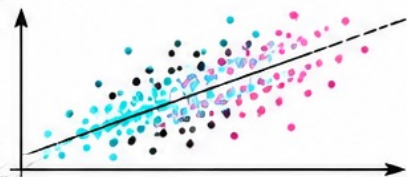
curl (X_C)



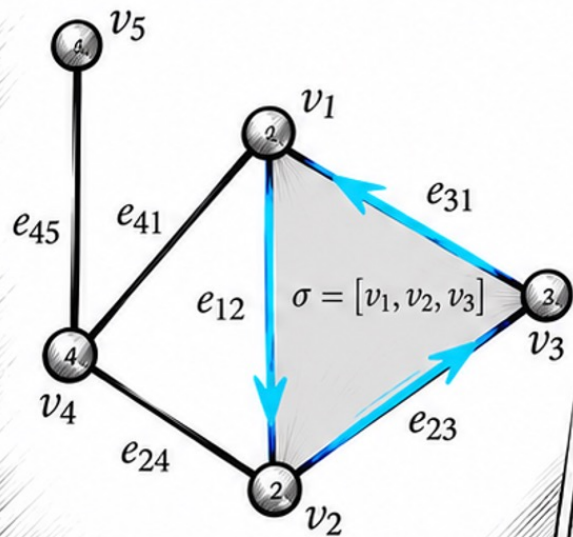
harmonic (X_H)

Inference & learning

Statistical inference
Prediction, classification



Simplicial complex



Boundary matrix B_2

(2-simplices $\rightarrow$ 1-simplices)

$$\begin{matrix} \sigma \\ e_{12} \\ e_{23} \\ e_{31} \\ e_{24} \\ e_{41} \\ e_{45} \end{matrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \partial\sigma = e_{12} + e_{23} + e_{31}$$

Coboundary matrix B_2^T

(1-simplices $\rightarrow$ 2-simplices)

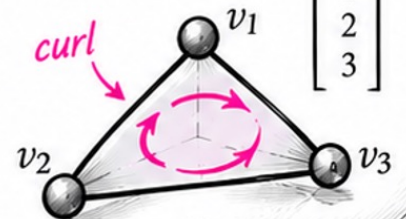
$$B_2^T = \begin{bmatrix} e_{12} & e_{23} & e_{31} & e_{24} & e_{41} & e_{45} \\ 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Edge signal $X \in \mathbb{R}^6$

$$\begin{matrix} e_{12} \\ e_{23} \\ e_{31} \\ e_{24} \\ e_{41} \\ e_{45} \end{matrix} \begin{bmatrix} 1 \\ 2 \\ 5 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

Face value (sum over edges)

$$B_2^T X = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 5 \\ 1 \\ 2 \\ 3 \end{bmatrix} = 8$$



Generalization to B_k

$$\begin{matrix} \dots & B_k & B_k^T \\ \begin{bmatrix} \pm 1 & 0 & \dots & 0 \\ 0 & \pm 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \pm 1 \end{bmatrix} & \begin{bmatrix} \pm 1 & 0 & \dots \\ 0 & \pm 1 & \dots \\ \vdots & \vdots & \ddots \\ 0 & 0 & \dots & \pm 1 \end{bmatrix} \\ \text{faces} \rightarrow (k-1)\text{-faces} & (k-1)\text{-faces} \rightarrow \text{faces} \\ \text{(boundary matrix)} & \text{(coboundary matrix)} \end{matrix}$$

k-th Hodge Laplacian

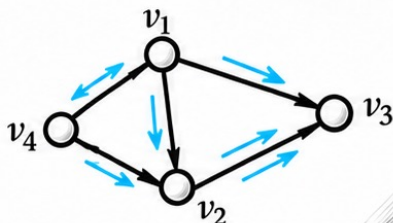
$$\Delta_1 = B_1 \overset{\substack{\# \text{ of nodes} \\ \uparrow}}{B_1^T} + \overset{\substack{\# \text{ of edges} \\ \uparrow}}{B_2} B_2^T$$

of edges # of faces

k = 0 : Graph Laplacian

$$\Delta_0 = B_1 B_1^T$$

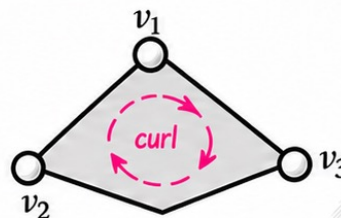
Acts on nodes
(0-simplices).
Measures
net outflow
at each node.



k = 1 : Edge Laplacian

$$\Delta_1 = B_1^T B_1 + B_2 B_2^T$$

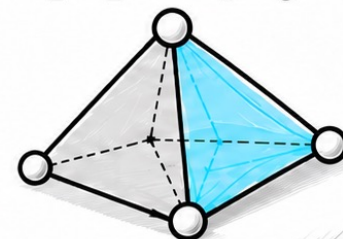
Acts on edges
(1-simplices).
Measures both
divergence and curl
of edge flows.



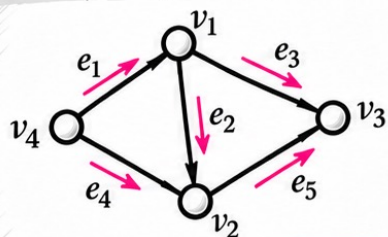
k = 2 : Face Laplacian

$$\Delta_2 = B_2^T B_2 + B_3 B_3^T$$

Acts on triangles
(2-simplices).
Measures how
face values are
inconsistent
across tetrahedra
(3-simplices).



Example (k = 1)



Edge $e_3 = (v_1 \rightarrow v_3)$ is included.

B_1 (edges $\times$ nodes)

	v_1	v_2	v_3	v_4
e_1	-1	0	0	1
e_2	1	-1	0	0
e_3	-1	0	1	0
e_4	0	-1	0	1
e_5	0	1	-1	0

Each column has -1 at the tail
and +1 at the head of the edge.

B_2 (faces $\times$ edges)

	e_1	e_2	e_3	e_4	e_5
f_1	1	1	-1	0	0

Triangle $f_1 = [v_1, v_2, v_3]$
has oriented boundary
 $\partial f_1 = e_1 + e_2 - e_3$.

Edge Laplacian matrix

$$\Delta_1 = B_1^T B_1 + B_2 B_2^T$$

	e_1	e_2	e_3	e_4	e_5
e_1	2	1	-1	0	0
e_2	1	2	-1	0	0
e_3	-1	-1	2	0	0
e_4	0	0	0	1	-1
e_5	0	0	0	-1	1

First 3 edges (e_1, e_2, e_3)
form a cycle (curl),
while e_4 and e_5 are tree
edges. The Hodge
Laplacian captures both
divergence and curl
structures.

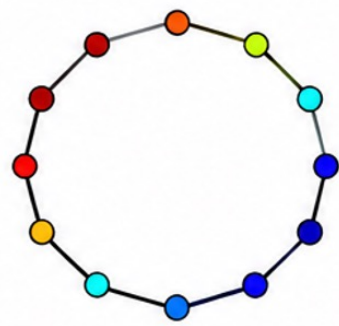
Eigenmodes of graph Laplacian vs. 1-Hodge Laplacian

Graph Laplacian (nodes)

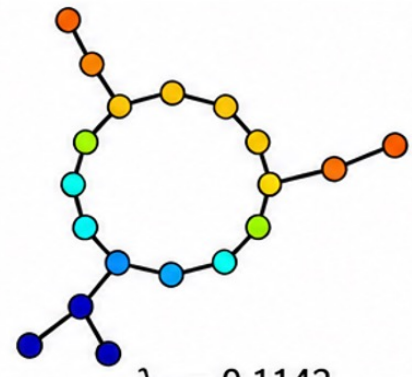
First nonzero eigenmode (λ_1)



$\lambda_1 = 0.0594$



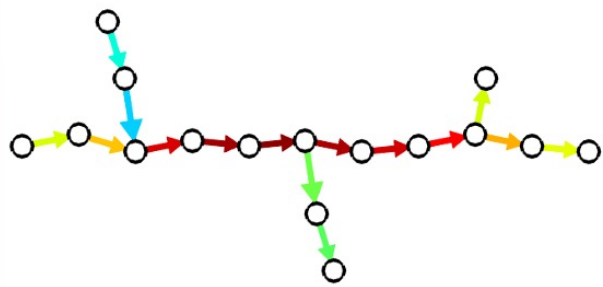
$\lambda_1 = 0.2679$



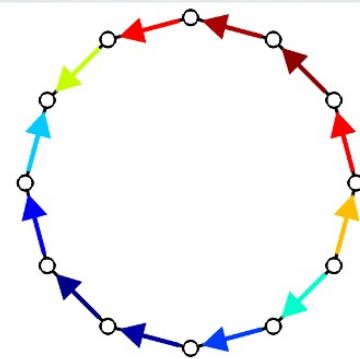
$\lambda_1 = 0.1142$

1-Hodge Laplacian (edges)

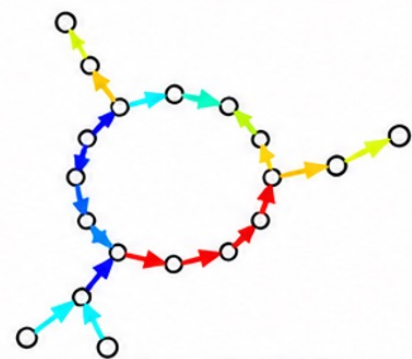
Gradient mode (λ_1)



$\lambda_1 = 0.0594$



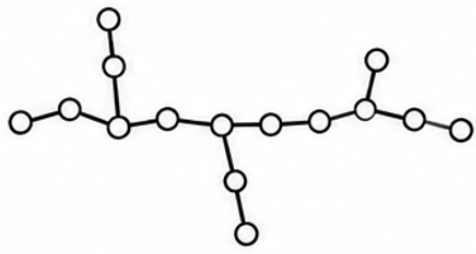
$\lambda_1 = 0.2679$



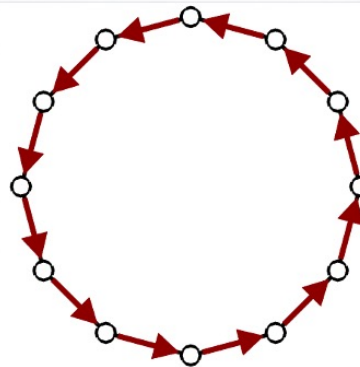
$\lambda_1 = 0.1142$

1-Hodge Laplacian (edges)

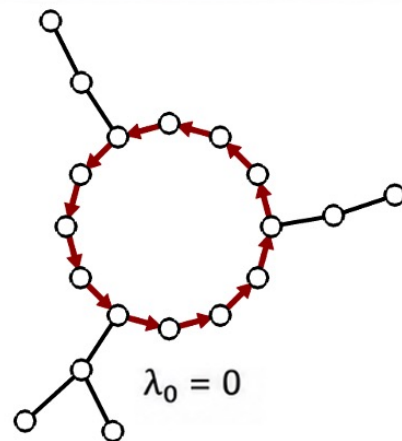
Harmonic mode ($\lambda_0 = 0$)



$\lambda_0 = 0$
(none)

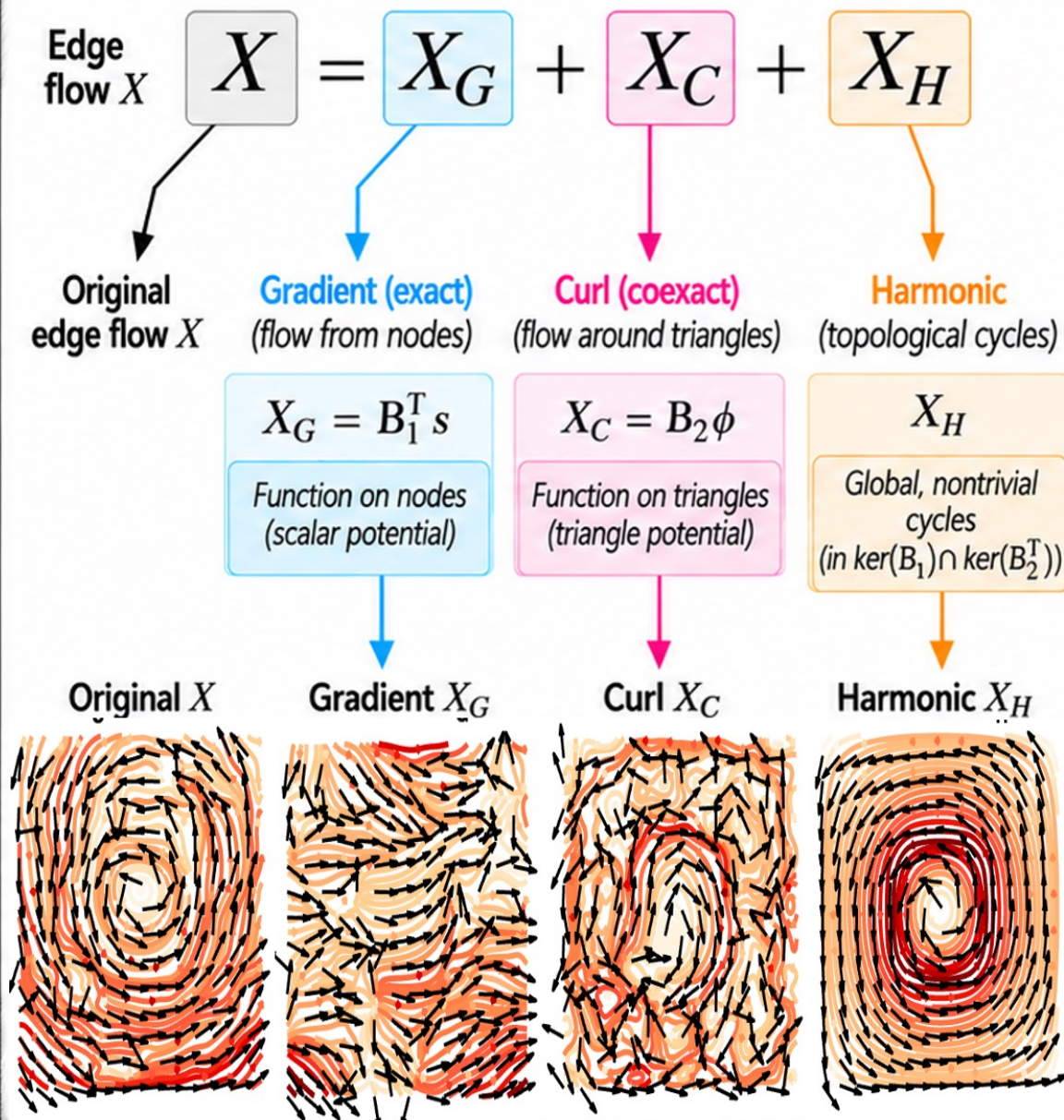


$\lambda_0 = 0$



$\lambda_0 = 0$

Hodge decomposition



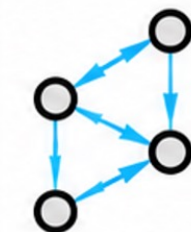
Least Squares Estimation of X_H

$$X_H = \arg \min_{Z \in \mathbb{R}^{[E]}} \|X - Z\|^2$$

subject to $B_1 Z = 0, \quad B_2^T Z = 0$

$B_1 Z = 0$
 (no divergence at nodes)

$B_2^T Z = 0$
 (no curl on triangles)



Zero net flow at each node

$$B = \begin{bmatrix} B_1 \\ B_2^T \end{bmatrix}$$



Zero circulation on each triangle

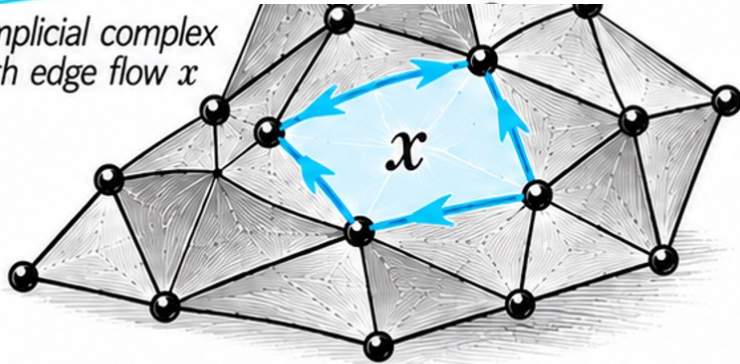
Orthogonal projection onto harmonic subspace

$$\mathcal{P}_H = I - B^T (BB^T)^+ B$$

$$X_H = \mathcal{P}_H X$$

From boundary matrix to cycles and harmonics

Simplicial complex
with edge flow x

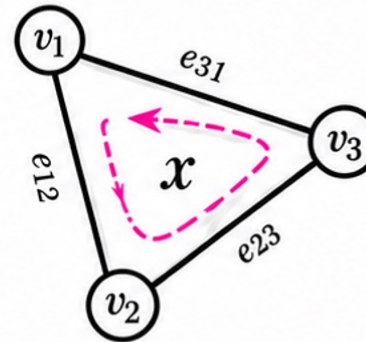


$$C_k = \ker(B_k) = \{x \in \mathbb{R}^{|E_k|} : B_k x = 0\}$$

k -cycles: edge flows with zero boundary
(vertex divergence)

1-cycles (loops)

$$C_1 = \ker(B_1)$$



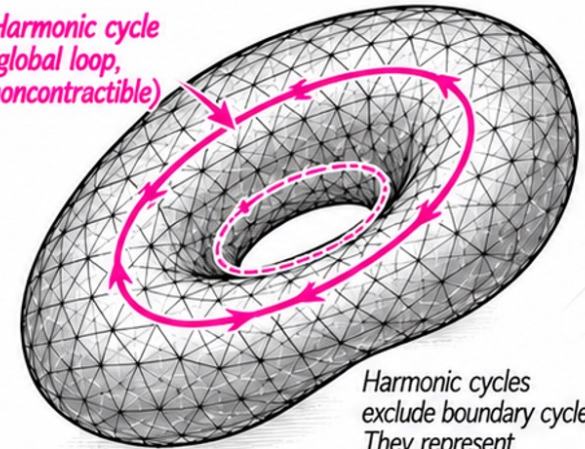
$$B_1 x = 0$$

(zero net flow at each node)

Harmonic subspace

$$H_k = \ker(\Delta_k) = \ker(B_k) \cap \ker(B_{k+1}^T)$$

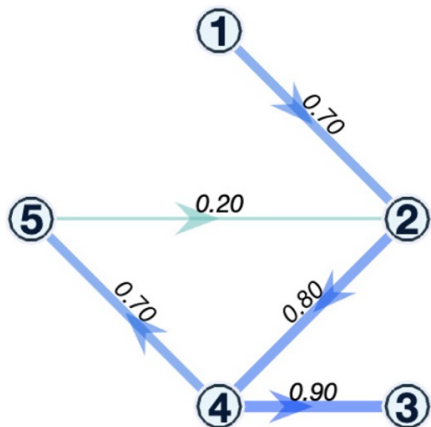
Harmonic cycle
(global loop,
noncontractible)



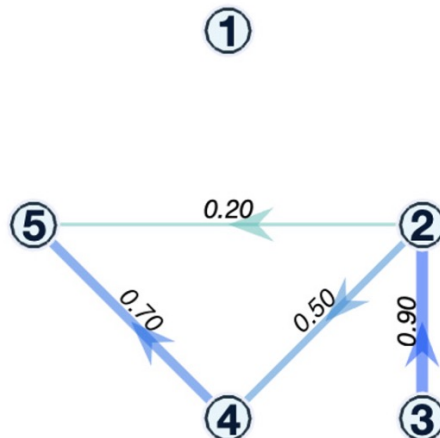
Harmonic cycles
exclude boundary cycles.
They represent
topologically nontrivial
loops (noncontractible).

$\ker(\Delta_1)$: Vector space of cycles

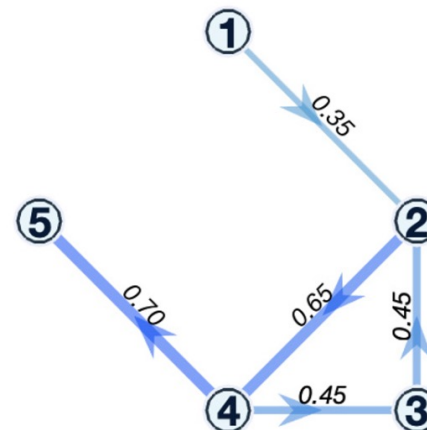
Network 1



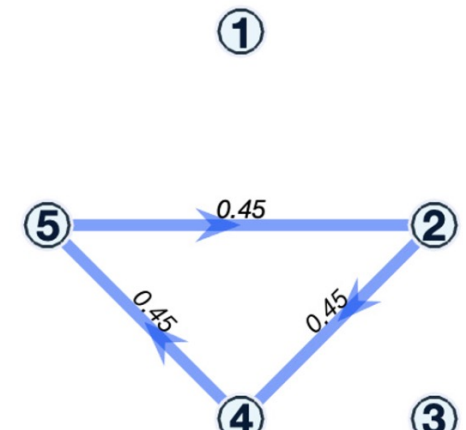
Network 2



Algebraic Averaging

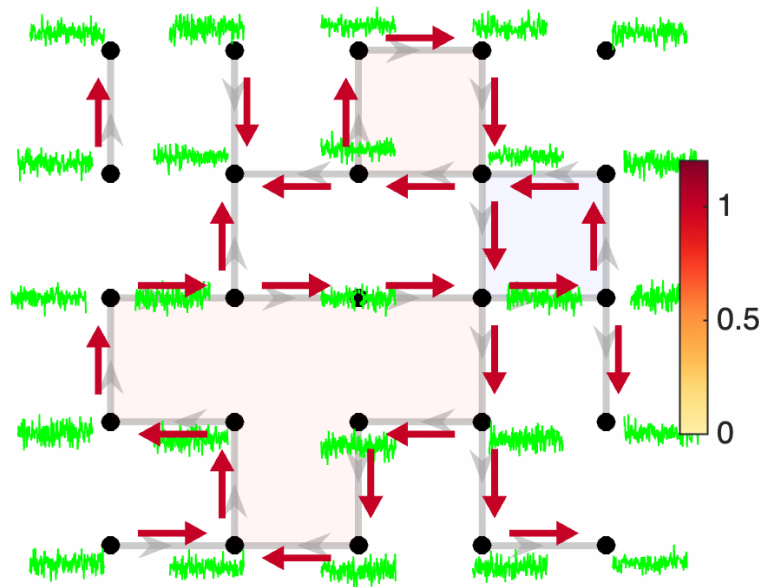


Averaging in Cycle Space



Cycles $C_1, C_2 \in \ker(\Delta_1) \longrightarrow \alpha_1 C_1 + \alpha_2 C_2 \in \ker(\Delta_1)$

Diffusion along edges



Diffusion equation on edges

$$\dot{X}(t) = -\Delta_1 X(t)$$

↓ Solution

$$X(t) = e^{-\Delta_1 t} X(0)$$

↓ Spectral expansion

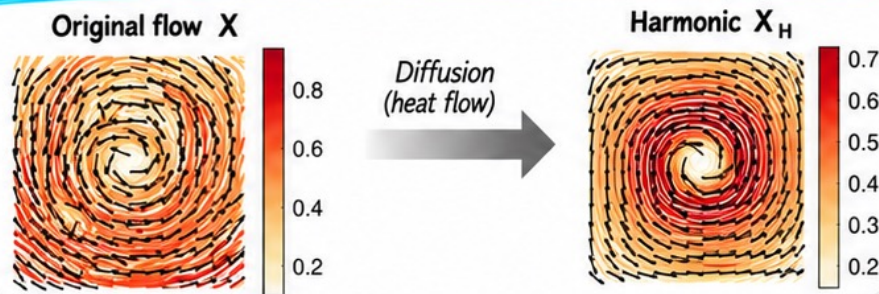
$$X(t) = \sum_k e^{-\lambda_k t} \psi_k \psi_k^\top X(0)$$

↓ $t \rightarrow \infty$

Harmonic cycle

$$X_H = \mathcal{P}_H X(0)$$

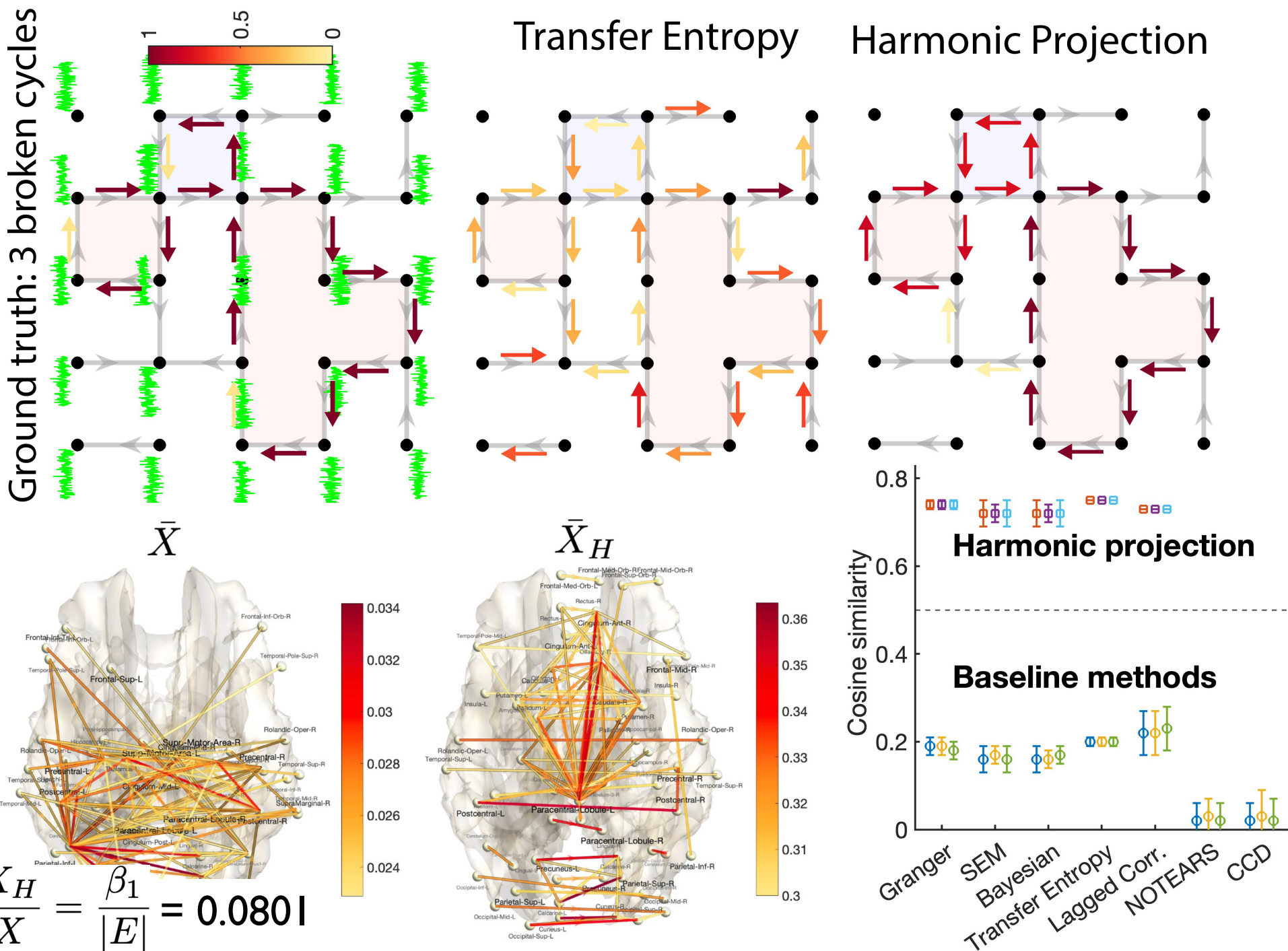
Intuition: diffusion removes non-harmonic flows



MATLAB code (Hodge 1-Laplacian)

```
[Phi, Lambda] = eig(full(Hodge1));
lambda = diag(Lambda);
c = Phi' * XO;
t = 2.0;
Xt = Phi * (exp(-t*lambda) .* c);
```

Performance in a cycle recovery task



Thank you

Any question, suggestion or collaboration
mkchung@wisc.edu